

A
Project Report

On
“PRISM”

SUBMITTED BY-
Rohit Bighot - 51991

UNDER THE GUIDANCE OF
Prof. Satish Kulkarni

Savitribai Phule Pune University
MASTER OF COMPUTER APPLICATION



JSPM's
JAYAWANTRAO SAWANT COLLEGE OF ENGINEERING
MCA Department
For Academic Year 2025-26



JAYAWANT SHIKSHAN PRASARAK MANDAL's

Jayawantrao Sawant College of Engineering

(Approved by AICTE, New Delhi, Govt of Maharashtra and Affiliated to University of Pune)

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FOUNDER SECRETARY

Prof.Dr. Pradeep Anandrao Patil

BE (Mechanical)

ME (Mechanical) Heat Power Engineering

Ph.D, (Mechanical) : HVAC & Heat Transfer

Principal

CERTIFICATE

This is to certify that **Mr. Rohit Gorakh Bighot** is a bonafide student of Jayawantrao Sawant College of Engineering, Hadapsar, Pune-28, has successfully completed the project work in the Partial Fulfillment of Master of Computer Application (MCA) Program as per guidelines provided by the Savitribai Phule Pune University for Academic Year 2025-26

Prof. Satish Kulkarni

Project Guide

Prof. Swayam Shah

HOD

Prof. Dr. Pradeep A Patil

Principal

External Examiner:

Date: / /2026

Internal Examiner:

Place: Pune



JAYAWANT SHIKSHAN PRAKASH MANDAL



Jayawantrao Sawant College of Engineering Hadapsar.

JSCOE Association For Developing Entrepreneurs

(Approved by Ministry of Corporate Affairs ,New Delhi India)

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Chairman

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ME (Mechanical) Heat Power Engineering
Ph.D. (Mechanical): HVAC & Heat Transfer
PRINCIPAL

EXPERIENCE LETTER

Dear **Rohit Gorakh Bighot**,

This is to certify that **Mr. Rohit Gorakh Bighot** has successfully completed her internship as a QA Intern at JADE (JSCOE Association for Developing Entrepreneurs, a Section 8 Company) from **26 Dec 2025 to 31 May 2026**.

During his internship, he contributed to the development, deployment, and maintenance of the organization's ERP system. He worked on **frontend development**, system architecture, production deployment using Docker, performance optimization, troubleshooting, and database management, helping deliver reliable and efficient software solutions.

Her key responsibilities included:

- Designing and developing responsive frontend applications and user interfaces
- Contributing to system architecture design and implementation
- Deploying and managing applications in production environments using Docker
- Optimizing website performance, reducing load times, and improving overall system efficiency
- Troubleshooting and resolving technical issues across the application stack
- Implementing and maintaining database backup and recovery procedures

Throughout the internship, **Mr. Rohit Gorakh Bighot** demonstrated strong analytical and problem-solving skills, a proactive attitude, and a willingness to learn. Her contributions were valuable to the team, and she maintained a professional and responsible approach towards her work.

We wish her all the best in her future endeavors.

Yours sincerely,

For JADE (JSCOE Association for Developing Entrepreneurs, a Section 8 Company)

Prof. Dr. Pradeep A Patil
PRINCIPAL

JADE (JSCOE Association for Developing Entrepreneurs, a Section 8 Company)

Declaration

To,
The Principal,
Jayawantrao Sawant college of Engineering,
Hadapsar, Pune-28

Respected Sir,

I, **Mr. Rohit Gorakh Bighot** hereby declare that the Project entitled “**PRISM**” has been developed and submitted under the guidance of Prof. Satish Kulkarni is my original work.

The Project work presented here is my own work and has not been duplicated from any other sources.

Date: / /2026
Place : Pune

Yours Sincerely
Rohit Gorakh Bighot

Acknowledgement

I would take this opportunity to thank Prof. Dr. Pradeep A Patil, Principal of JSCOE for their extended support to me throughout the course.

I would like to thank Prof. Swayam shah sir, HOD, Department of Computer Applications for scholarly disposition, timely guidance, support and cooperation.

I would like to thank Prof. Satish Kulkarni for his kind guidance, keen interest, continuous encouragement and inspiration throughout the project work.

Finally, I am grateful to all the staff members of JSCOE for their cooperation and support.

I also thankful to get constant encouragement, support and guidance from all teaching and non-teaching staff for their support which helped us in successfully completion of our project work.

Mr. Rohit Gorakh Bighot

Exam Seat No: 51991

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1. INTRODUCTION

1.1 Project Profile

In modern educational institutions, managing academic and administrative activities through manual or fragmented systems leads to inefficiency, data inconsistency, and poor accessibility. Colleges handle large volumes of student data, academic records, assessments, attendance, and documentation, which require a secure and well-structured digital solution.

A College Enterprise Resource Planning (ERP) system provides an integrated platform that centralizes institutional data and streamlines academic and administrative workflows. By implementing role-based access control and automated processes, an ERP system improves accuracy, transparency, and operational efficiency.

The College ERP Management System is designed to address these challenges by offering a secure, scalable, and role-driven web application. The system emphasizes centralized control, strict authorization, and semester-wise academic lifecycle management, ensuring that critical operations are performed only by authorized users while maintaining data integrity and security.

1.2 Existing System

The existing system suffers from:

- Manual or semi-digital student record handling
- No centralized academic tracking (CCE, T&P, projects, activities)
- Poor access control (data editable by unauthorized users)
- Difficulty in tracking semester-wise performance
- Fragmented document storage

There is no single platform that integrates **attendance, academic records, certifications, and placements** securely.

1.3 Scope of Project

The scope of the project includes:

- JWT-based authentication and authorization
- Role-based academic and administrative control
- Semester-wise academic tracking
- Centralized document management
- Department-level access for HODs
- Subject-level access for teachers

The system **strictly avoids student self-registration**, ensuring institutional control over data integrity.

1.4 Operating Environment (H/W & S/W)

Hardware

- Processor: Intel i5 or higher
- RAM: 8 GB minimum
- Storage: 100 GB

Software

- OS: Windows / Linux
 - Backend: Java, Spring Boot
 - Security: Spring Security + JWT
 - Database: PostgreSQL
 - Frontend: React.js
 - Tools: IntelliJ IDEA, Git, Postman
-

1.5 Technology & Tools Used

- **Spring Boot:** Backend framework
 - **JWT (JSON Web Token):** Stateless authentication
 - **Spring Security:** Role-based authorization
 - **PostgreSQL:** Relational database
 - **REST APIs:** Client-server communication
-

2. PROPOSED SYSTEM

2.1 Proposed System

The proposed system is a **secure ERP platform** where:

- Admin has complete system authority
- HOD manages department-specific data
- Teachers manage subject-specific academic records
- Students can view their academic and activity data

The system enforces **strict authorization rules** using JWT tokens and role-based access control.

2.2 Modules of Proposed System

- Authentication & Authorization Module
- Student Management Module
- Academic Management Module
- Attendance Management Module
- Document & Activity Management Module

2.2.1 Detailed Description of Proposed Modules

1. Authentication & Authorization Module

- JWT token-based authentication
- Stateless session management
- Role-based access (ADMIN, HOD, TEACHER, STUDENT)
- Secure token validation for every request
- Prevents unauthorized API access

No JWT = No access. Period.

2. Student Management Module

- Student creation **only by Admin or HOD**
 - No student self-registration
 - Student profile management
 - Edit student details (Admin/HOD)
 - Department and semester mapping
 - View-only access for students
-

3. Academic Management Module

- Semester-wise academic tracking
 - CCE marks entry
 - Training & Placement record management
 - Competition participation tracking
 - Field project and activity project details
 - Subject-wise marks management
 - Marks entry allowed only for assigned teachers
-

4. Attendance Management Module

- Subject-wise attendance tracking
 - Semester-based attendance records
 - Attendance entry by assigned teachers
 - View access for students
-

5. Document & Activity Management Module

- Upload certificates and documents
 - Competition proof uploads
 - Project submission documents
 - Document viewing by Admin, HOD, and teachers
 - Secure file storage and access control
-

2.3 Objective of Proposed System

- Enforce secure academic data management
 - Maintain data integrity through controlled access
 - Enable semester-wise student performance tracking
 - Reduce manual paperwork
 - Provide a scalable ERP foundation
-

2.4 Fact Finding Technique

- Interviews with Admin, HOD, and faculty
 - Analysis of current academic workflows
 - Observation of attendance and assessment processes
 - Review of existing ERP limitations
-

2.5 Feasibility Study

Technical Feasibility

- Uses mature and industry-standard technologies
- JWT ensures scalable authentication

Operational Feasibility

- Role-based UI simplifies user interaction
- Minimal training required

Economic Feasibility

- Open-source frameworks
 - Low deployment and maintenance cost
-

3. ANALYSIS AND DESIGN

3.1 System Requirements

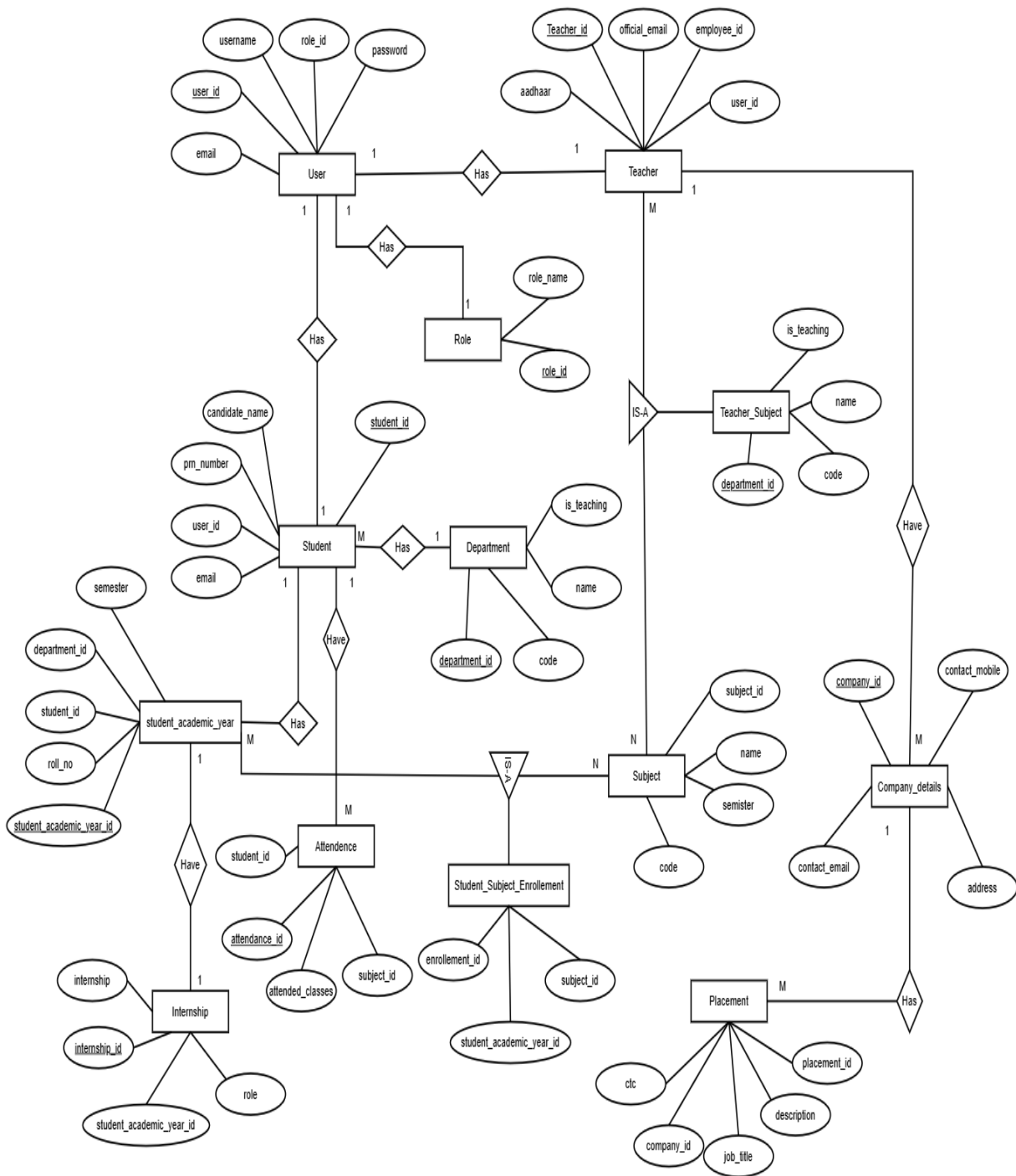
Functional Requirements

- JWT-based authentication
 - Role and permission management
 - Student data creation and modification
 - Subject-wise marks and attendance entry
 - Document upload and verification
 - Semester-wise academic reporting
-

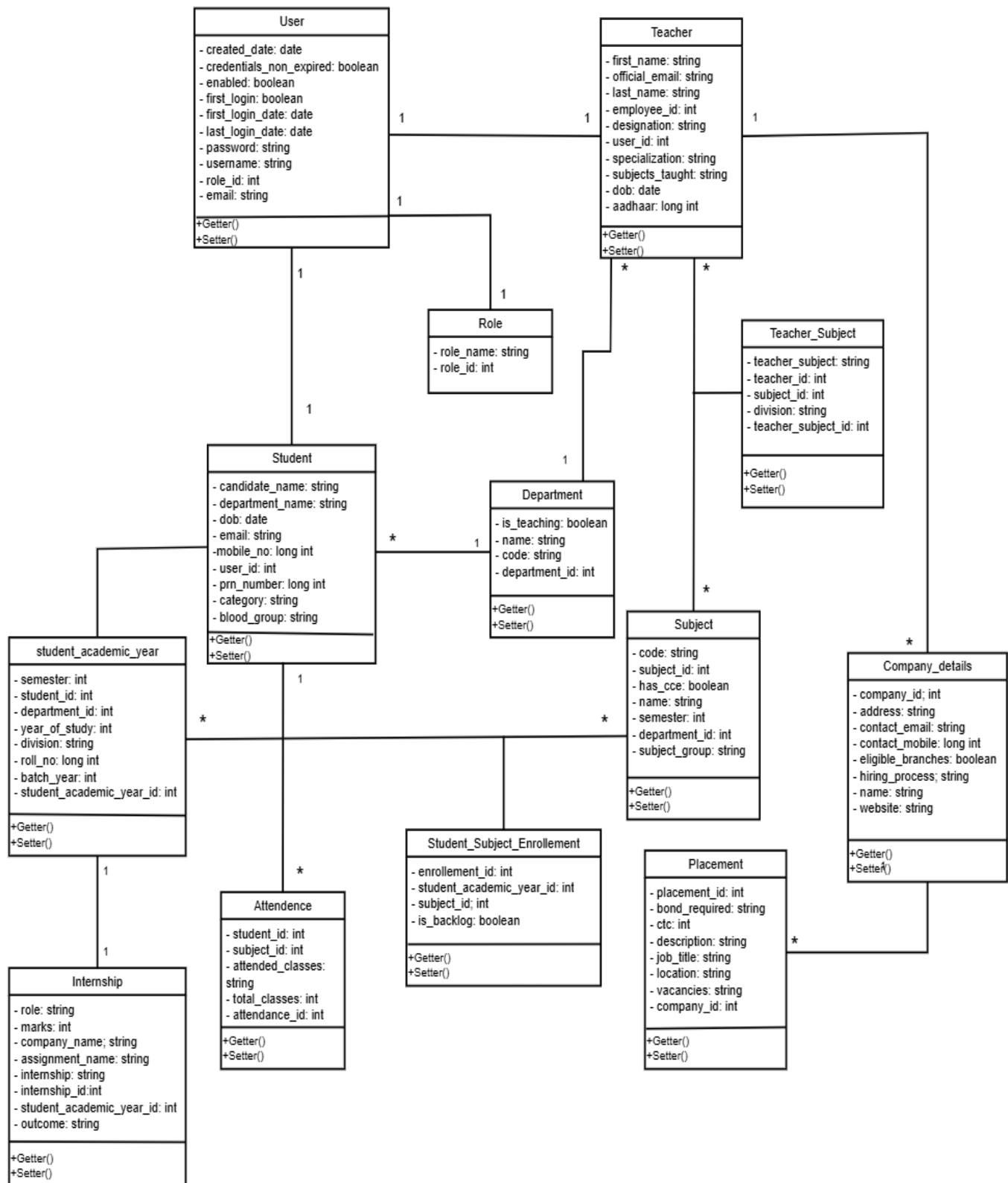
Non-Functional Requirements

- Security: Token-based authorization
- Performance: Fast API response
- Scalability: Modular architecture
- Reliability: Data consistency
- Maintainability: Clean layered

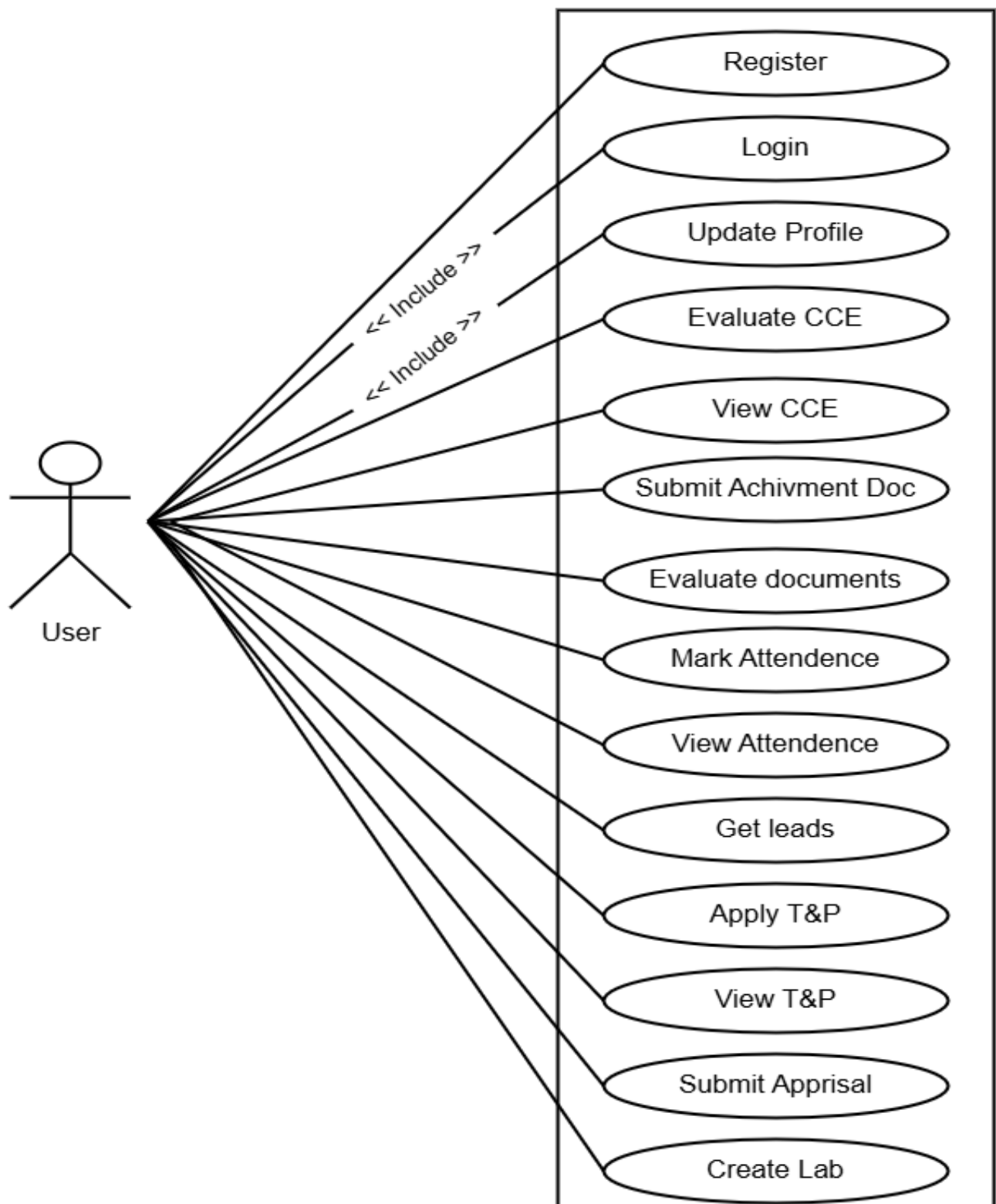
3.2) Entity Relationship Diagram



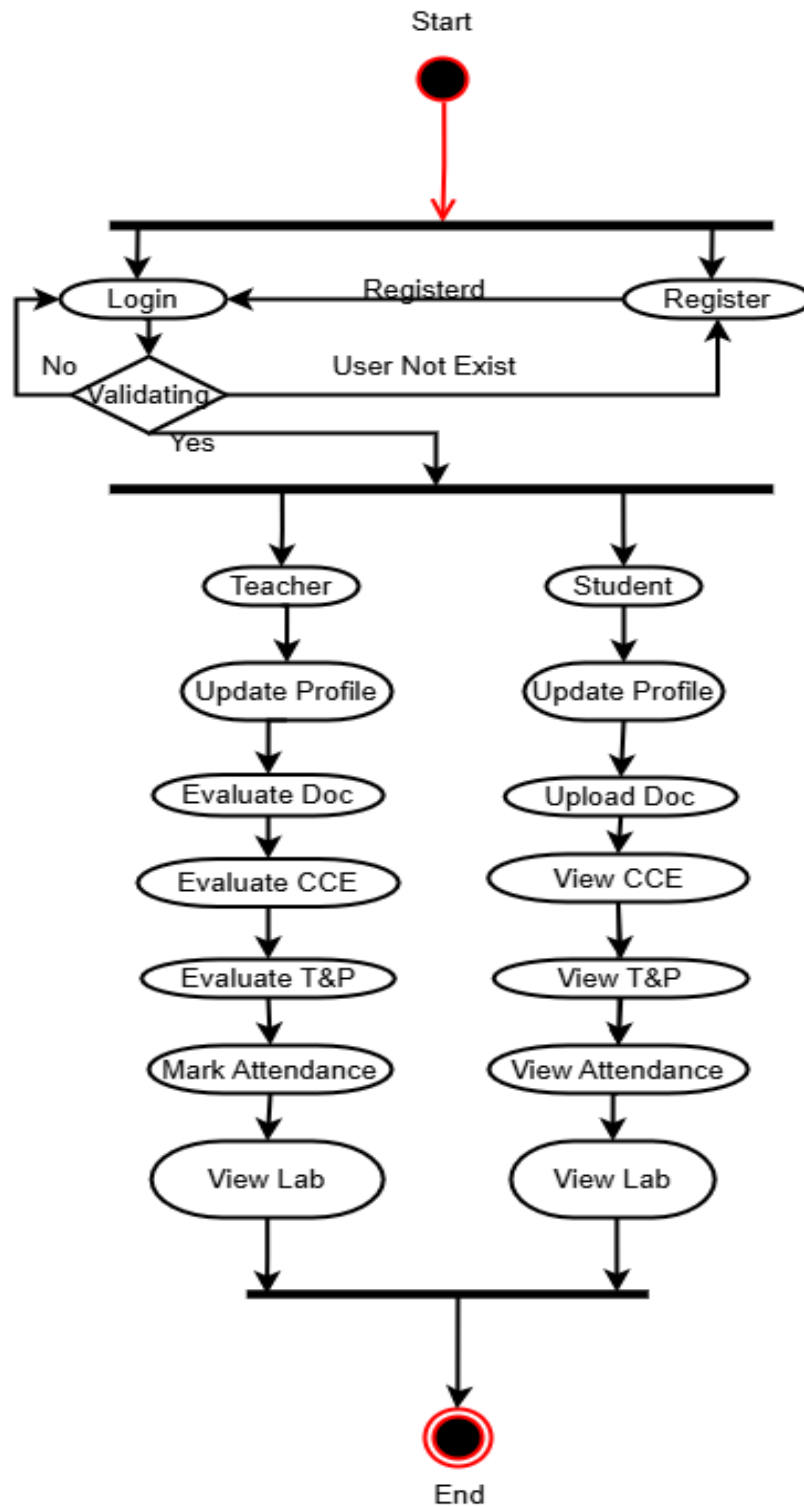
3.3.1) Class Diagram



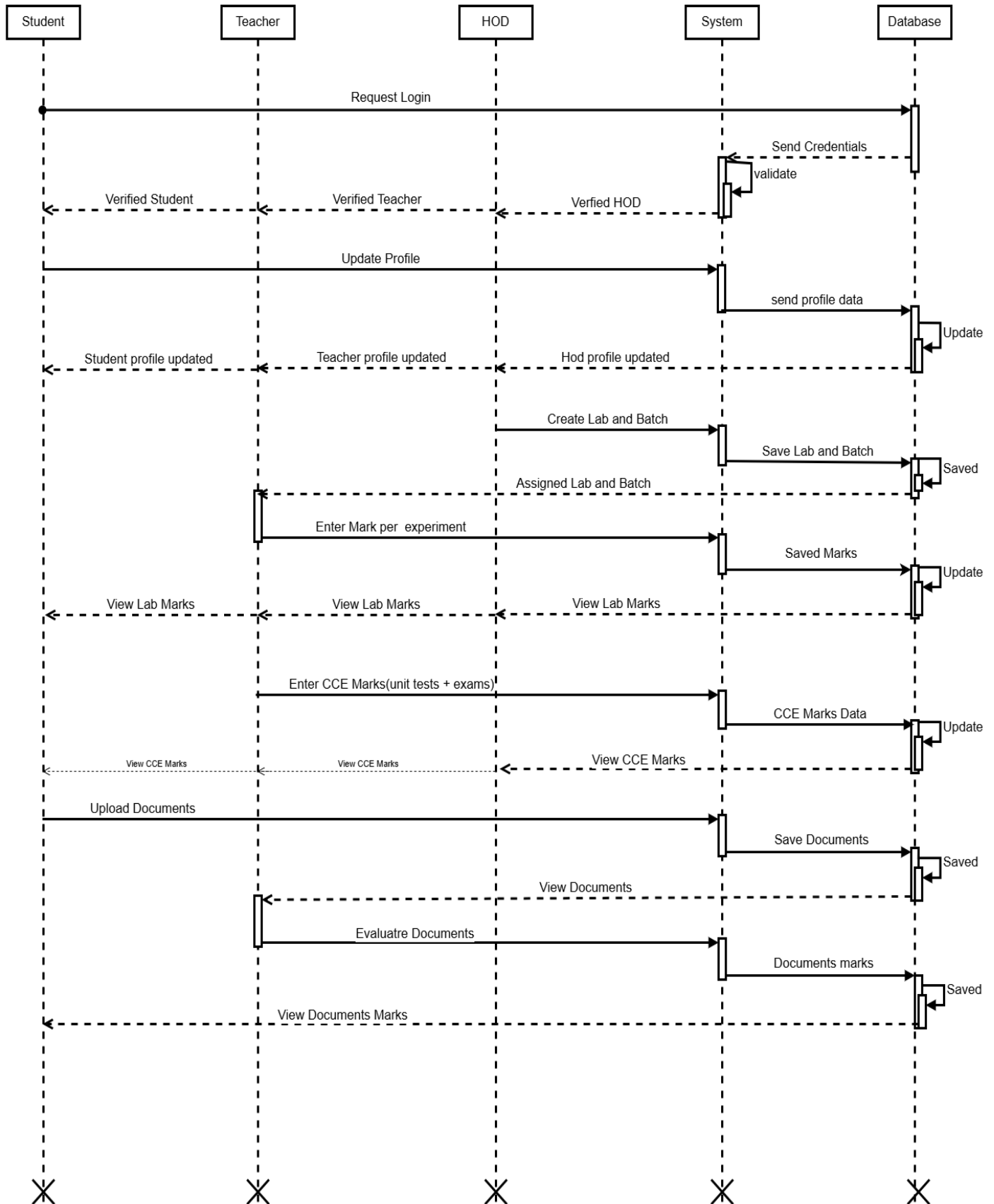
3.3.2 Use Case Diagram



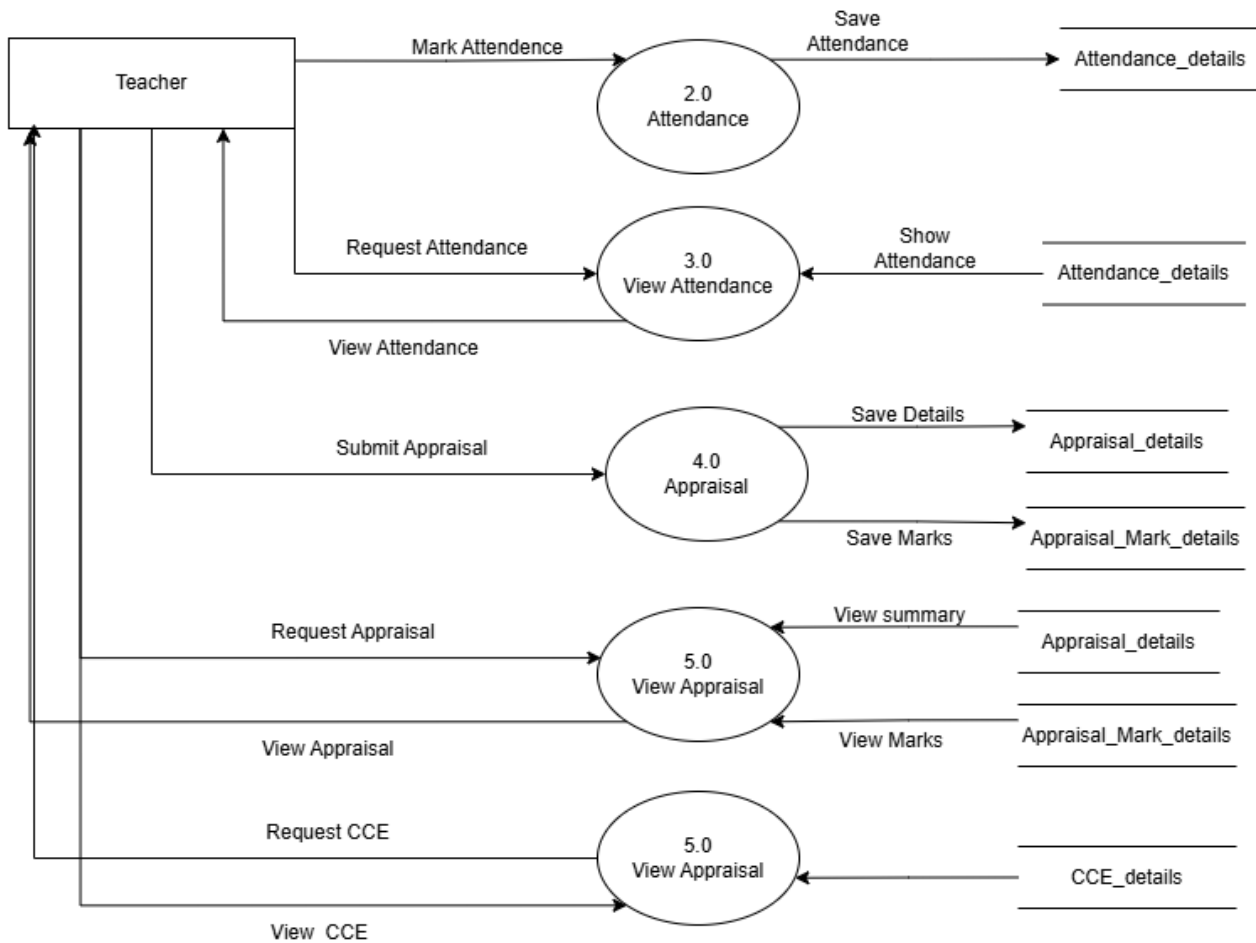
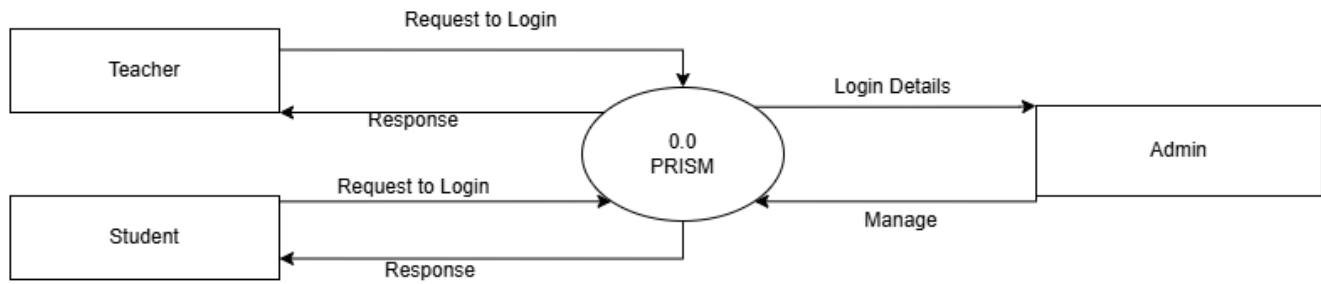
3.3.3) Activity Diagram

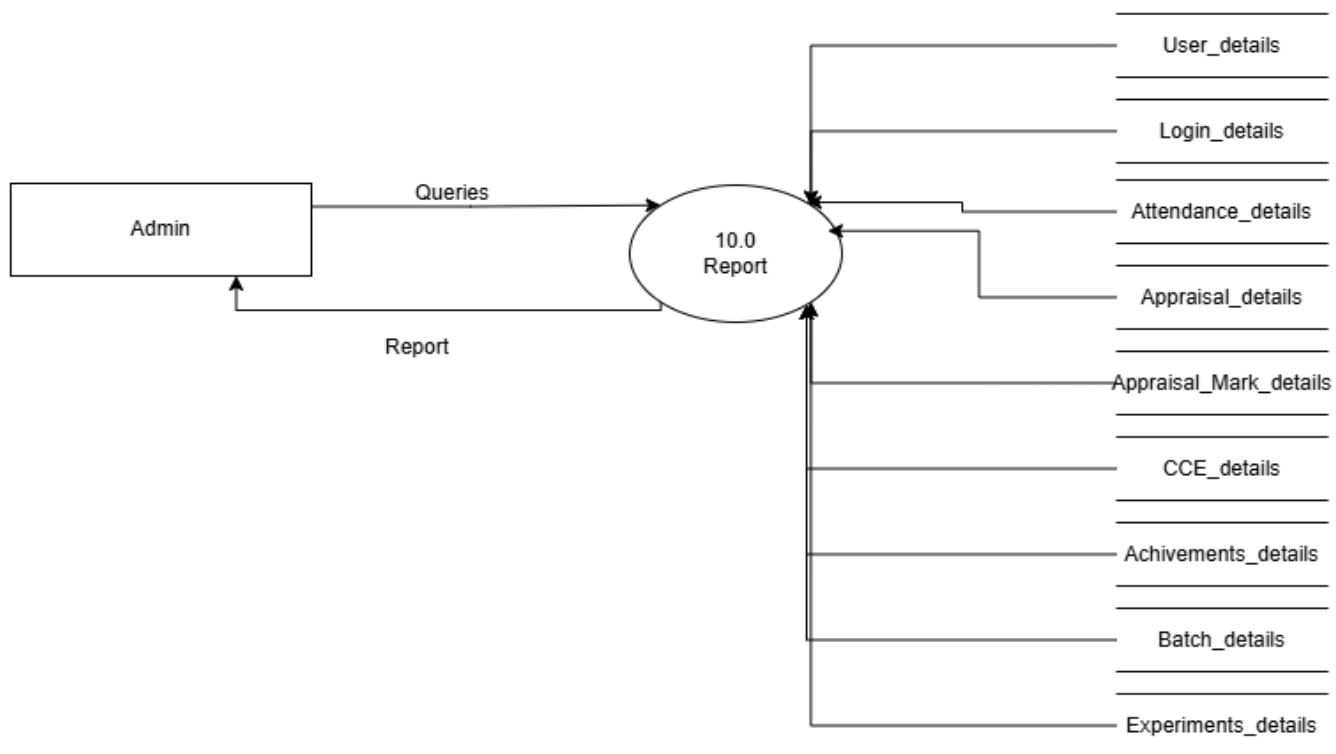
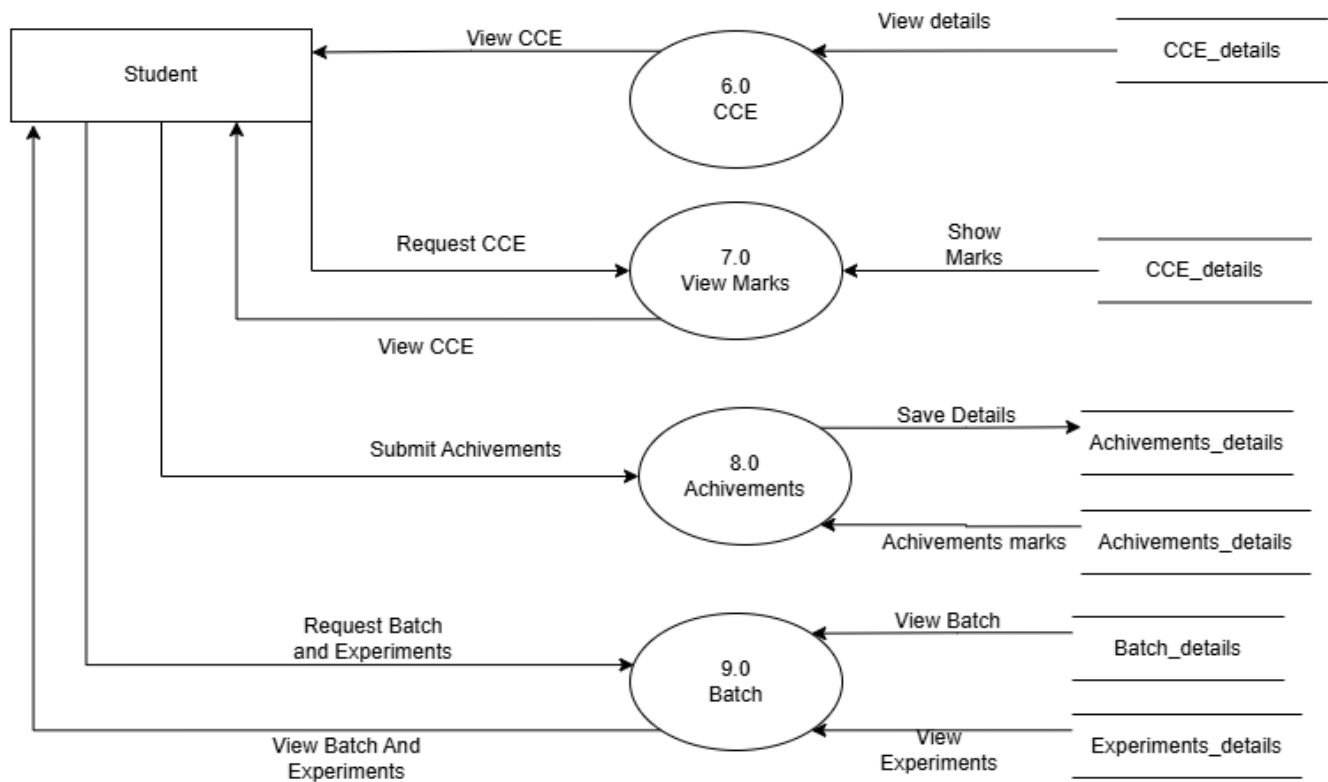


3.3.4) Sequence Diagram

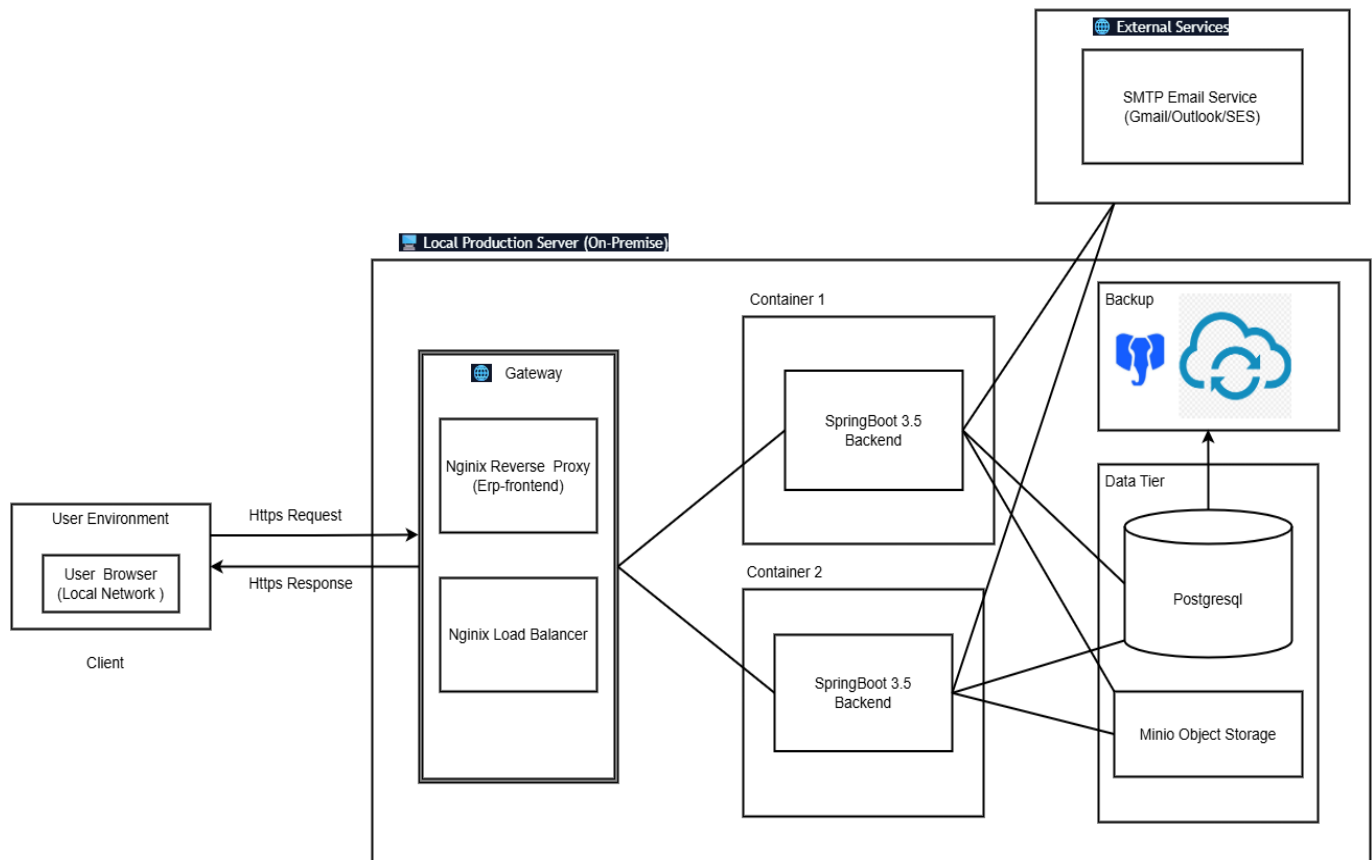


3.3.5) Data Flow Diagram (DFD)





3.3.6) Deploy Diagram



3.4) Table Structure

1. Users

Column Name	Data Type	Description
User_Id	BIGINT (PK)	Unique Identifier For Each User.
Username	VARCHAR(100)	Unique Username Used For Login.
Email	VARCHAR(100)	User's Email Address.
Password	VARCHAR(120)	Encrypted Password Of The User.
Account_Non_Locked	BOOLEAN	Indicates Whether The Account Is Locked.
Account_Non_Expired	BOOLEAN	Indicates Whether The Account Is Expired.
Credentials_Non_Expired	BOOLEAN	Indicates Whether The Credentials Are Expired.
Enabled	BOOLEAN	Indicates Whether The Account Is Enabled.
Role_Id	BIGINT (FOREIGN KEY)	Reference To Role Table Defining User Permissions.
Attributes	ONETOONE RELATIONSHIP	Additional User Attributes Linked To This User.
Password_Reset_Tokens	ONETOMANY RELATIONSHIP	List Of Password Reset Tokens Associated With The User.
First_Login	BOOLEAN (DEFAULT: TRUE)	Indicates Whether This Is The User's First Login.
Created_Date	TIMESTAMP (AUTO GENERATED)	Timestamp When The User Account Was Created.
First_Login_Date	TIMESTAMP	Timestamp Of The User's First Login.
Last_Login_Date	TIMESTAMP	Timestamp Of The User's Last Login.
Student	ONETOONE RELATIONSHIP	Linked Student Entity (If User Is A Student).
Teacher	ONETOONE RELATIONSHIP	Linked Teacher Entity (If User Is A Teacher).

2. Student

Column Name	Data Type	Description
Student Id	BIGINT (PK)	Unique Identifier For Each Student.
User_Id	BIGINT (FK)	Reference To User Table For Authentication And Account Details.
Candidate_Name	VARCHAR	Full Candidate Name As Per Admission Form.
First_Name	VARCHAR	Student's First Name.
Middle Name	VARCHAR	Student's Middle Name.
Last_Name	VARCHAR	Student's Last Name.
Mobile_No	VARCHAR	Student's Mobile Contact Number.
Prn_Number	VARCHAR(20)	Permanent Registration Number Assigned To The Student.
Email	VARCHAR	Student's Email Address.

Gender	ENUM	Gender Of The Student (Stored As String).
Dob	DATE	Date Of Birth Of The Student.
Religion	VARCHAR	Religion Of The Student.
Category	VARCHAR	Category (E.G., General, Obc, Sc, St, Etc.).
Profile_Picture_Path	VARCHAR	File Path Of The Student's Profile Picture.
Blood_Group	ENUM (STRING)	Blood Group Of The Student (Stored As String).
Addresses	ONETOMANY RELATIONSHIP	List Of Addresses Associated With The Student.
Student_Academic_Years	ONETOMANY RELATIONSHIP	Academic Year Details Linked To The Student.
Previous_Education	ONETOONE RELATIONSHIP	Previous Education Details Of The Student.
Student_Interests	ONETOONE RELATIONSHIP	Student Interests Information.
Student_Identity	ONETOONE RELATIONSHIP	Identity Details Of The Student.
Skills	ONETOMANY RELATIONSHIP	Skills Associated With The Student.

3. Student Academic Year

Column Name	Data Type	Description
Student_Academic_Year_Id	BIGINT (PK)	Unique ID for student academic year record
Student_Id	BIGINT (FK)	References student record
Department_Id	BIGINT (FK)	References department record
Year_Of_Study	INTEGER	Stores year of study
Semester	INTEGER	Stores semester number
Batch_Year	VARCHAR	Stores admission batch/academic year
Division	VARCHAR	Stores student division
Roll_No	INTEGER, UNIQUE	Stores roll number
Semester_Start_Date	DATE	Stores semester start date
Semester_End_Date	DATE	Stores semester end date
Is_Active	BOOLEAN (DEFAULT: TRUE)	Indicates active academic record status
Certificates	ONETOMANY RELATIONSHIP	Tracks earned certificates
Internships	ONETOMANY RELATIONSHIP	Tracks internship records
Mini_Projects	ONETOMANY	Tracks mini project records

	RELATIONSHIP	
Field_Projects	ONETOMANY RELATIONSHIP	Tracks field project records
Competitions	ONETOMANY RELATIONSHIP	Tracks competition participation records
Enrollments	ONETOMANY RELATIONSHIP	Tracks subject enrollment records

4. Department

Column Name	Data Type	Description
Department_Id	BIGINT (PK)	Unique ID for department
Name	VARCHAR(100)	Stores department name
Code	VARCHAR(20)	Stores unique department code
Is_Teaching	BOOLEAN	Indicates teaching department status
Subjects	ONETOMANY RELATIONSHIP	Tracks offered subjects
Student_Academic_Years	ONETOMANY RELATIONSHIP	Tracks student academic year records
Teachers	ONETOMANY RELATIONSHIP	Tracks associated teachers

5. Subject

Column Name	Data Type	Description
Subject_Id	BIGINT (PK)	Unique ID for subject
Department_Id	BIGINT (FK)	References department record
Name	VARCHAR(100)	Stores subject name
Subject_Group	VARCHAR	Stores subject group classification
Has_Cce	BOOLEAN	Indicates CCE applicability
Code	VARCHAR(50)	Stores unique subject code
Theory_Credits	INTEGER (DEFAULT: 0)	Stores theory credit count
Year_Of_Study	INTEGER	Stores applicable study year
Semester	INTEGER	Stores offered semester
Subject_Category	ENUM (STRING)	Stores subject category
Elective_Group	INTEGER	Stores elective group ID
Total_Units	INTEGER	Stores total syllabus units
Teacher_Subjects	ONETOMANY RELATIONSHIP	Maps assigned teachers

Status	ENUM (STRING, DEFAULT: ACTIVE)	Stores subject status
Created_At	TIMESTAMP (AUTO)	Stores record creation timestamp
Updated_At	TIMESTAMP (AUTO)	Stores last updated timestamp
Labs	ONETOMANY RELATIONSHIP	Tracks associated lab sessions
Enrollments	ONETOMANY RELATIONSHIP	Tracks student enrollments

6. Student Subject Enrollment

Column Name	Data Type	Description
Id	BIGINT (PK)	Unique ID for student subject enrollment
Student_Academic_Year_Id	BIGINT (FK)	References student academic year record
Subject_Id	BIGINT (FK)	References subject record
Is_Backlog	BOOLEAN (DEFAULT: FALSE)	Indicates backlog subject status
Enrollment_Date	TIMESTAMP	Stores enrollment date and time
Status	ENUM (STRING, DEFAULT: CONFIRMED)	Stores enrollment status
Exams	ONETOMANY RELATIONSHIP	Tracks linked student exam records
Unit_Assessments	ONETOMANY RELATIONSHIP	Tracks linked unit assessment records

7. Teacher Subject

Column Name	Data Type	Description
Teacher_Subject_Id	BIGINT	Unique ID for teacher-subject assignment
Teacher_Id	BIGINT	References teacher entity
Subject_Id	BIGINT	References subject entity
Division	VARCHAR(5)	Stores class section/division code
Is_Active	BOOLEAN	Indicates whether assignment is active

8. Company

Column Name	Data Type	Description
Company_Id	BIGINT (PK)	Unique ID for company record
Name	VARCHAR	Stores organization name
Address	VARCHAR(1000)	Stores company address
Website	TEXT	Stores company website URL
Hr_Name	VARCHAR	Stores HR contact person name
Contact_Email	TEXT	Stores official contact email
Contact_Mobile	TEXT	Stores official contact number
Eligible_Branches	TEXT	Stores eligible hiring branches
Hiring_Process	TEXT	Stores hiring process details

9. Internship

Column Name	Data Type	Description
Internshipid	BIGINT(PK)	Unique ID for internship record
Company_Name	VARCHAR	Stores internship company name
Role	VARCHAR	Stores student role/designation
Assignment_Name	TEXT	Stores project or assignment details
Outcome	TEXT	Stores internship outcome/learning result
Start_Date	DATE	Stores internship start date
End_Date	DATE	Stores internship end date
Minio_Object_Key	VARCHAR	Stores unique MinIO certificate file path
Student_Academic_Year_Id	BIGINT	References student academic year mapping
Marks	DOUBLE	Stores internship evaluation marks

10. Placement

Column Name	Data Type	Description
Placementid	BIGINT (PK)	unique ID for placement record
Company_Id	BIGINT (FK)	References registered company
Job_Title	VARCHAR	Stores job role/designation
Description	VARCHAR(1000)	Stores job description and responsibilities
Location	VARCHAR	Stores work location(s)
Registration_Start_Date	DATE	Stores application start date
Registration_End_Date	DATE	Stores application deadline
Drive_Start_Date	DATE	Stores recruitment drive start date
Ctc	NUMERIC(12,2)	Stores CTC/package details
Bond_Required	VARCHAR	Stores bond/service agreement details
Vacancies	INT	Stores vacancy count
Status	VARCHAR(20)	Stores posting workflow status
Referred_By_Teacher_Id	BIGINT	References initiating teacher
Placement_Students	JOIN TABLE	Maps student relationships
Applications	JOIN TABLE	Tracks placement applications

3.5) Data Dictionary

1. User

Column Name	Data Type	Constraint	Description
User_Id	BIGINT	Pk, Auto Increment	Unique Identifier For Each User.
Username	VARCHAR(100)	Unique, Not Null	Username Used For Login.
Email	VARCHAR(100)	Unique, Not Null	User Email Address.
Password	VARCHAR(120)	Not Null	Encrypted Password.
Account_Non_Locked	BOOLEAN	Default: True	Indicates Whether Account Is Locked.
Account_Non_Expired	BOOLEAN	Default: True	Indicates Whether Account Is Expired.
Credentials_Non_Expired	BOOLEAN	Default: True	Indicates Whether Credentials Are Expired.
Enabled	BOOLEAN	Default: True	Indicates Whether Account Is Enabled.
Role_Id	BIGINT	Fk	Reference To Role Table.
Attributes	ONETOONE	Relationship	Additional User Attributes.
Password_Reset_Tokens	ONETOMANY	Relationship	Password Reset Tokens Linked To User.
First_Login	BOOLEAN	Default: True	Indicates First Login.
Created_Date	TIMESTAMP	Auto Generated	Account Creation Timestamp.
First_Login_Date	TIMESTAMP	Nullable	First Login Timestamp.
Last_Login_Date	TIMESTAMP	Nullable	Last Login Timestamp.
Student	ONETOONE	Relationship	Linked Student Entity.
Teacher	ONETOONE	Relationship	Linked Teacher Entity.

2. Students

Column Name	Data Type	Constraint	Description
Student_Id	BIGINT	Pk, Auto Increment	Unique Student Identifier.
User_Id	BIGINT	Fk, Unique, Not Null	Reference To User Table.
Candidate_Name	VARCHAR	Nullable	Full Name As Per Admission Form.
First_Name	VARCHAR	Not Null	First Name.

Middle_Name	VARCHAR	Nullable	Middle Name.
Last_Name	VARCHAR	Not Null	Last Name.
Mobile_No	VARCHAR	Nullable	Mobile Number.
Prn_Number	VARCHAR(20)	Unique	Permanent Registration Number.
Email	VARCHAR	Nullable	Student Email.
Gender	ENUM (STRING)	Nullable	Gender.
Dob	DATE	Nullable	Date Of Birth.
Religion	VARCHAR	Nullable	Religion.
Category	VARCHAR	Nullable	Category.
Profile_Picture_Path	VARCHAR	Nullable	Profile Picture Path.
Blood_Group	ENUM (STRING)	Nullable	Blood Group.
Addresses	ONETOMANY	Relationship	Student Addresses.
Student_Academic_Years	ONETOMANY	Relationship	Academic Year Records.
Previous_Education	ONETOONE	Relationship	Previous Education Details.
Student_Interests	ONETOONE	Relationship	Student Interests.
Student_Identity	ONETOONE	Relationship	Identity Details.
Skills	ONETOMANY	Relationship	Student Skills.

3. Student Academic Year

Column Name	Data Type	Constraint	Description
Student_Academic_Year_Id	BIGINT	Pk, Auto Increment	Unique Academic Year Record Id.
Student_Id	BIGINT	Fk, Not Null	Reference To Student.
Department_Id	BIGINT	Fk, Not Null	Reference To Department.
Year_Of_Study	INTEGER	Not Null	Year Of Study.
Semester	INTEGER	Not Null	Semester Number.
Batch_Year	VARCHAR	Nullable	Admission Batch Year.
Division	VARCHAR	Nullable	Division.
Roll_No	INTEGER	Unique	Roll Number.
Semester_Start_Date	DATE	Nullable	Semester Start Date.
Semester_End_Date	DATE	Nullable	Semester End Date.
Is_Active	BOOLEAN	Default: True	Indicates Active Record.
Certificates	ONETOMANY	Relationship	Certificates Earned.
Internships	ONETOMANY	Relationship	Internships.
Mini_Projects	ONETOMANY	Relationship	Mini Projects.

Field_Projects	ONETOMANY	Relationship	Field Projects.
Competitions	ONETOMANY	Relationship	Competitions.
Enrollments	ONETOMANY	Relationship	Subject Enrollments.

4. Department

Column Name	Data Type	Constraint	Description
Department_Id	BIGINT	Pk, Auto Increment	Unique Department Id.
Name	VARCHAR(100)	Not Null	Department Name.
Code	VARCHAR(20)	Unique, Not Null	Department Code.
Is_Teaching	BOOLEAN	Not Null	Indicates Teaching Department.
Subjects	ONETOMANY	Relationship	Subjects Offered.
Student_Academic_Years	ONETOMANY	Relationship	Linked Academic Year Records.
Teachers	ONETOMANY	Relationship	Department Teachers.

5. Subject

Column Name	Data Type	Constraint	Description
Subject_Id	BIGINT	Pk, Auto Increment	Unique Subject Id.
Department_Id	BIGINT	Fk, Not Null	Reference To Department.
Name	VARCHAR(100)	Not Null	Subject Name.
Subject_Group	VARCHAR	Nullable	Subject Group.
Has_Cce	BOOLEAN	Nullable	Cce Enabled.
Code	VARCHAR(50)	Unique, Not Null	Subject Code.
Theory_Credits	INTEGER	Default: 0	Theory Credits.
Year_Of_Study	INTEGER	Not Null	Applicable Year.
Semester	INTEGER	Not Null	Semester.
Subject_Category	ENUM (STRING)	Not Null	Subject Category.
Elective_Group	INTEGER	Not Null	Elective Group Id.
Total_Units	INTEGER	Nullable	Total Units.
Teacher_Subjects	ONETOMANY	Relationship	Teacher Assignments.
Status	ENUM (STRING)	Default: Active	Subject Status.
Created_At	TIMESTAMP	Auto Generated	Creation Timestamp.
Updated_At	TIMESTAMP	Auto Updated	Last Updated Timestamp.
Labs	ONETOMANY	Relationship	Associated Labs.
Enrollments	ONETOMANY	Relationship	Subject Enrollments.

6. Student Subject Enrollment

Column Name	Data Type	Constraint	Description
Id	BIGINT	Pk, Auto Increment	Enrollment Id.
Student_Academic_Year_Id	BIGINT	Fk, Not Null	Reference To Student Academic Year.
Subject_Id	BIGINT	Fk, Not Null	Reference To Subject.
Is_Backlog	BOOLEAN	Default: False	Indicates Backlog Subject.
Enrollment_Date	TIMESTAMP	Auto Generated	Enrollment Timestamp.
Status	ENUM (STRING)	Default: Confirmed	Enrollment Status.
Exams	ONETOMANY	Relationship	Student Exams.
Unit_Assessments	ONETOMANY	Relationship	Unit Assessments.

7. Teacher Subject

Column Name	Data Type	Constraints	Description
Teacher_Subject_Id	BIGINT	Primary Key, Auto Incre	Unique ID for teacher-subject assignment
Teacher_Id	BIGINT	Not Null, Foreign Key (Many-To-One)	References the assigned teacher
Subject_Id	BIGINT	Not Null, Foreign Key (Many-To-One)	References the assigned subject
Division	VARCHAR(5)	Max_Length = 5	Stores class section/division code
Is_Active	BOOLEAN	Default True	Indicates whether the assignment is active

8. Company

Column Name	Data Type	Constraints	Description
Company_Id	BIGINT	Primary Key, Auto_Increment	Unique ID for company record (auto-incremented)
Name	VARCHAR	Not Null	Stores company/organization name
Address	VARCHAR	Max_Length = 1000	Stores company address
Website	TEXT	None	Stores company website URL
Hr_Name	VARCHAR	None	Stores HR contact person name
Contact_Email	TEXT	None	Stores official contact email

Contact_Mobile	TEXT	None	Stores official contact number
Eligible_Branches	TEXT	Converted (List To String)	Stores eligible academic branches list

9. Internship

Column Name	Data Type	Constraints	Description
Internshipid	BIGINT	Primary Key, Auto_Increment	Unique ID for internship record (auto-incremented)
Company_Name	VARCHAR	Not Null	Stores internship company name
Role	VARCHAR	None	Stores student role/designation
Assignment_Name	TEXT	None	Stores project or assignment details
Outcome	TEXT	None	Stores internship outcome/learning result
Start_Date	DATE	None	Stores internship start date
End_Date	DATE	None	Stores internship end date
Minio_Object_Key	VARCHAR	Not Null, Unique	Stores MinIO certificate file path
Student_Academic_Year_Id	BIGINT	Not Null, Foreign Key (Many-To-One)	References student academic year mapping
Marks	DOUBLE	Default 0.0	Stores internship evaluation marks

10. Placement

Column Name	Data Type	Constraints	Description
Placementid	BIGINT	Primary Key, Auto_Increment	Unique ID for placement record
Company_Id	BIGINT	Not Null, Foreign Key	References registered company
Job_Title	VARCHAR	Not Null	Stores job role/designation
Description	VARCHAR(1000)	Max_Length = 1000	Stores job description and responsibilities
Location	VARCHAR	None	Stores job location(s)

Registration_Start_Date	DATE	None	Stores application start date
Registration_End_Date	DATE	None	Stores application deadline
Drive_Start_Date	DATE	None	Stores recruitment drive start date
Ctc	NUMERIC(12,2)	Precision = 12, Scale = 2	Stores CTC/package details
Bond_Required	VARCHAR	None	Stores bond/service agreement details
Vacancies	INT	None	Stores total vacancies/openings
Status	VARCHAR(20)	Not Null, Max_Length = 20, Default 'Draft'	Stores current posting status
Referred_By_Teacher_Id	BIGINT	Foreign Key	References initiating teacher
Placement_Students	JOIN TABLE	Many-To-Many Relationship	Maps many-to-many student relationships
Applications	JOIN TABLE	One-To-Many (Mappedby)	Tracks student application records

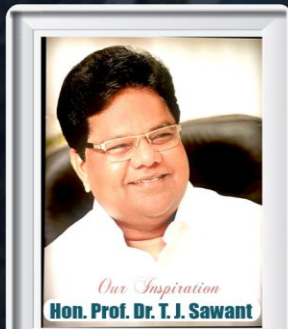
3.5 Sample Input and Output Screens

1. Input (Login page)



JSPM's
Jayawantrao Sawant College of Engineering, Hadapsar, Pune.
(Approved By AICTE & Affiliated To Savitribai Phule Pune University. & Accredited with A+ Grade by NAAC)





Our Inspiration
Hon. Prof. Dr. T. J. Sawant

Founder Secretary- JSPM Group of Institutes, Pune.
President- TSSM Group of Institutes, Pune.
President- SBERCT Group of Institutes, Barshi

JSPM - PRISM @ JSCOE

(Performance Review & Institutional Success Model)

A centralized platform to manage students, teachers, academics, and administration—designed to simplify college operations and improve communication across JSPM.

Student Management

Track attendance, results, profiles, and academic progress in real time.

Faculty & Admin Control

Manage teachers, subjects, schedules, notices, and internal workflows securely.

This system is developed and maintained by our dedicated contributors and technical team.

Contributors

Welcome back

Sign in to continue.

Username

rb0090120@gmail.com

Password

Password@123

☐ Remember me for 30 days

Sign In

Forgot your password?

Don't have an account? Create Account

Output (Login Page)

JSPM PRISM
TEACHER

Profile

Attendance

Show Attendance

Student

Industry Connect

Principal Office Letters

Teacher Appraisal

Feedback

Logout



JSPM's
Jayawantrao Sawant College of Engineering, Hadapsar, Pune.
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Mr. Dummy For Test
PROFESSOR
Artificial Intelligence and Data Science
ACCEPTED

EMAIL

rb0090120@gmail.com

PHONE

0000000000

EMPLOYEE ID

0000

Overview

Academic

Experience

Personal Information

Gender

Male

Date of Birth

2026-01-01

Address

Pune

demo, demo, demo

demo - 000000

2. CCE Dashboard (Marks are entered as input)

e.g Student Name : Pooja Bhagat

JSPM PRISM
ADMIN

Teacher

student

Student Dashboard

Training & Placement

CCE

submission

Subject Enrollment

Batch

Industry Collaboration

Feedback

Logout

JSPM's

Jayawantrao Sawant College of Engineering, Hadapsar, Pune.

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Department

Batch

Semester

Division

Subject

Experiential learning

Internal test

CCE Marks

STUDENT INFO		UNIT 1		UNIT 2		UNIT 3		UNIT 4		UNIT 5		TOTAL		OTHER
ROLL NO	NAME	FIELD (0-10)	QUIZ (0-10)	FIELD (0-10)	QUIZ (0-10)	FIELD (0-10)	QUIZ (0-10)	FIELD (0-10)	QUIZ (0-10)	FIELD (0-10)	QUIZ (0-10)	TOTAL QUIZ	TOTAL FIELD ACTIVITY	ATTENDANCE 100%
2101	KAKADE SANIKA VIJAYKUMAR	7.5	10	8.5	8	9.5	9	8	9	8	9	45	41.5	89.66
2102	BHAGAT POOJA VIJAY	9	10	9	9	8.5	9	9.5	8	9.5	8	44	45.5	93.1
2103	Kadam Akansha Raghunath	9	10	9	9	9	8	9.5	8	9.5	9	44	46	92
2104	RATHOD VAISHNAVI VISHNU	9	8	9.5	8	8.5	8	9.5	8	9.5	8	40	46	82.76
2105	JADHAV SANJANA DATTATRAY	9	10	9.5	9	9.5	9	9.5	8	9.5	8	44	47	96.55
2106	GAWAI ATHANG ASHOK	9.5	10	9	8	9.5	8	9.5	8	9.5	8	42	47	68.97
2107	SAGAIKAR RISHITA RAHUL	9.5	10	9	8	7.5	8	9.5	8	9.5	8	42	45	82.76
2108	YADAV SANCHITA NITIN	9	10	9.5	8	9.5	9	9.5	8	9.5	8	43	47	96.55
2109	SHERE SANCHITA SANTOSH	9	10	9	0	9.5	0	9.5	0	9.5	0	10	46.5	96.55
2110	PANDIT KHUSHI	0.5	10	0.5	0	0.5	0	0	0	0	0	15	15.5	03.1

JSPM PRISM
ADMIN

Teacher

student

Student Dashboard

Training & Placement

CCE

submission

Subject Enrollment

Batch

Industry Collaboration

Feedback

Logout

JSPM's

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Department

Batch

Semester

Division

Subject

Experiential learning

Internal test

CCE Marks

STUDENT INFO		INTERNAL TEST	
ROLL NO	NAME	MID TERM (0-40)	END TERM (0-60)
2101	KAKADE SANIKA VIJAYKUMAR	18	30
2102	BHAGAT POOJA VIJAY	27	32
2103	Kadam Akansha Raghunath	28	47
2104	RATHOD VAISHNAVI VISHNU	25	35
2105	JADHAV SANJANA DATTATRAY	30	37
2106	GAWAI ATHANG ASHOK	29	50
2107	SAGAIKAR RISHITA RAHUL	28	29
2108	YADAV SANCHITA NITIN	30	43
2109	SHERE SANCHITA SANTOSH	28	33
2110	PANDIT KHUSHI RATNAKAR	32	50
2111	KAWADE SANIKA SUNIL	27	39
2112	DESHPANDE MAITREE MUKUND	26	50
2113	PATIL ROSHANI MAHENDRA	28	40
2114	DESHMUKH ATHARVA KALYAN	26	26

JSPM PRISM
ADMIN

Teacher

student

Student Dashboard

Training & Placement

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Subject Enrollment

Batch

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Logout

JSPM's
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Department

Batch

Semester

Division

Subject

Electronics and Telec

2025-26

Semester 4

Division A

Communication Eng

Experiential learning

Internal test

CCE Marks

STUDENT INFO		30 MARKS DISTRIBUTION				
ROLL NO	NAME	FIELD (15)	QUIZ (5)	INTERNAL (5)	ATTENDANCE (0-5)	CCE MARKS (30)
2101	KAKADE SANIKA VIJAYKUMAR	12.45	4.50	2.40	4.48	23.83
2102	BHAGAT POOJA VIJAY	13.65	4.40	2.95	4.65	25.66
2103	Kadam Akansha Raghunath	13.80	4.40	3.75	4.60	26.55
2104	RATHOD VAISHNAVI VISHNU	13.80	4.00	3.00	4.14	24.94
2105	JADHAV SANJANA DATTATRAY	14.10	4.40	3.35	4.83	26.68
2106	GAWAI ATHANG ASHOK	14.10	4.20	3.95	3.45	25.70
2107	SAGAIKAR RISHITA RAHUL	13.50	4.20	2.85	4.14	24.69
2108	YADAV SANCHITA NITIN	14.10	4.30	3.65	4.83	26.88
2109	SHERE SANCHITA SANTOSH	13.95	1.00	3.05	4.83	22.83
2110	PANDIT KHUSHI RATNAKAR	13.65	4.60	4.10	4.65	27.01
2111	KAWADE SANIKA SUNIL	14.10	4.20	3.30	4.48	26.08
2112	DESHPANDE MAITREE MUKUND	14.10	4.70	3.80	4.65	27.25

Output

JSPM PRISM
STUDENT

Profile

Academic

Training and Placement

Submission

Attendance

CCE

Labs

Skills

Career & Placement

Feedback

Logout

JSPM's
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Subject Wise Marks (View Only)

Experiential Learning

Internal Test

CCE Marks

Subject Info	Unit 1		Unit 2		Unit 3		Unit 4		Unit 5		Internal test		Attendance
	Field Activity	Pedagogy Quiz	Field Activity	Pedagogy Quiz	Field Activity	Pedagogy Quiz	Field Activity	Pedagogy Quiz	Field Activity	Pedagogy Quiz	Mid Term	End Term	Percentage (%)
Communication Engineering	9	10	9	9	8.5	9	9.5	8	9.5	8	27	32	93.1
Control Systems	7.5	7	6	8	6.5	0	8.5	9	10	10	35	45	95.12
Environment Awareness	7.5	9	9	6	8	8	7.8	8	0	0	11	26	68.75
Object-Oriented Programming	7.5	8	7.5	10	9	10	9	9	9	9.33	21	47	84.85
Open Elective-II (Project Management)	8	10	8	10	7.5	10	8	10	0	0	14	22	69.23
Signals and Systems	8	10	9	8.67	9	0	8	0	8.5	0	36	41	92.11



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Subject Wise Marks (View Only)

Experiential Learning

Internal Test

CCE Marks

Subject Info		Internal test		Total Quiz and Activity	
Subject		Mid Term	End Term	Total Quiz	Total Activity
Communication Engineering		27	32	44	45.5
Control Systems		35	45	34	38.5
Environment Awareness		11	26	31	32.3
Object-Oriented Programming		21	47	46.33	42
Open Elective-II (Project Management)		14	22	40	31.5
Signals and Systems		36	41	18.67	42.5



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Subject Wise Marks (View Only)

Experiential Learning

Internal Test

CCE Marks

Subject Info	30 Marks Distribution				
Subject	Field Activity (15)	Quiz (5)	Internal test (5)	Attendance (5)	CCE Marks (30)
Communication Engineering	13.65	4.40	2.95	4.65	25.66
Control Systems	11.55	3.40	4.00	4.76	23.71
Environment Awareness	9.69	3.10	1.85	3.44	18.08
Object-Oriented Programming	12.60	4.63	3.40	4.24	24.88
Open Elective-II (Project Management)	9.45	4.00	1.80	3.46	18.71
Signals and Systems	12.75	1.87	3.85	4.61	23.07

Input:-

JSPM's Jayawantrao Sawant College of Engineering, Hadapsar, Pune.

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Student Portal > My Profile > Edit Profile

Output:-

[illegible]

4. Training and Placement Dashboard

Input:

JSPM PRISM
ADMIN

Teacher

student

Student Dashboard

Training & Placement

CCE

submission

Subject Enrollment

Batch

Industry Collaboration

Feedback

Logout

JSPM's
Jayawantrao Sawant College of Engineering, Hadapsar, Pune.
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Training & Placement Management System
Comprehensive student performance tracking and assessment

All changes savedDownload Excel

DepartmentBatchSemesterDivisionEmployability (30%)Technical Competency (50%)

STUDENT INFORMATION			MERIT ASSESSMENT (20%)	EMPLOYABILITY SKILLS (30%)								
ROLL NO	STUDENT NAME	DIV	SGPA SCORE (20 MARKS)	APTITUDE TESTS (MINIMUM 5 REQUIRED)					SOFT SKILLS TESTS (MINIMUM 3 REQUIRED)			SOFT SKILLS ATTENDANCE
				AT1	AT2	AT3	AT4	AT5	ST1	ST2	ST3	
2101	KAKADE SANIKA VIJAYKUMAR	A	7.5	8.67	9	9	7.67	0	0	0	0	0 %
2102	BHAGAT POOJA VIJAY	A	0	4.33	3	6.33	9	0	0	0	0	0 %
2103	Kadam Akansha Raghunath	A	0	7.33	7.33	7.67	0	0	0	0	0	0 %
2104	RATHOD VAISHNAVI VISHNU	A	0	9.33	7	10	8.67	0	0	0	0	0 %
2105	JADHAV SANJANA DATTATRAY	A	0	5	3.67	8	7.67	0	0	0	0	0 %
2106	GAWAI ATHANG ASHOK	A	0	8.67	9.67	9.67	8.67	0	0	0	0	0 %
2107	SAGAJKAR RISHITA RAHUL	A	0	9.33	7.33	0	7.33	0	0	0	0	0 %

JSPM PRISM
ADMIN

Teacher

student

Student Dashboard

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JSPM's
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Training & Placement Management System
Comprehensive student performance tracking and assessment

All changes savedDownload Excel

DepartmentBatchSemesterDivisionEmployability (30%)Technical Competency (50%)

STUDENT INFORMATION			TECHNICAL COMPETENCY ASSESSMENT (50%)						TOTAL SCORE
ROLL NO	STUDENT NAME	DIV	CERTIFICATIONS	SUBJECT QUIZZES	INTERNSHIP	COURSE PROJECT	NATIONAL EVENTS		
2101	KAKADE SANIKA VIJAYKUMAR	A	3.33	8.4	0	8.77	0	34.87	
2102	BHAGAT POOJA VIJAY	A	3	8.56	9	9.29	9	43.38	
2103	Kadam Akansha Raghunath	A	0	8.57	0	9.49	0	22.53	
2104	RATHOD VAISHNAVI VISHNU	A	0	8.81	0	8.79	0	24.60	
2105	JADHAV SANJANA DATTATRAY	A	0	6.48	0	8.89	0	20.24	
2106	GAWAI ATHANG ASHOK	A	1	8.75	0	9.5	0	26.59	
2107	SAGAJKAR RISHITA RAHUL	A	0	10.03	0	9.69	0	24.52	
2108	YADAV SANCHITA NITIN	A	0	10.37	0	9.93	0	25.83	
2109	SHERE SANCHITA SANTOSH	A	0	7.31	0	9.11	0	22.49	
2110	PANDIT KHUSHI RATNAKAR	A	3.5	10.49	0	9.7	0	30.89	

Output :

JSPM PRISM
STUDENT

Profile

Academic

Training and Placement

Submission

Attendance

CCE

Labs

Skills

Career & Placement

Industry Dashboard

Student Opportunities

My Applications

Feedback

Logout

JSPM's
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Student Training and Placement Report

Student Details

Name: BHAGAT POOJA VIJAY
Roll No: 2102
Division: A

Performance Overview (Total: 100 Marks)

Category	Score	Max
SPPU Result (SGPA)	0	20
Employability (30 Marks)		
Aptitude Test (Out of 10)	4.53	10
Softskill Test (Out of 10)	0.00	10
Softskill Attendance	0%	10
Technical Competency (50 Marks)		
Certification Courses	3	10
Subject Quiz	8.56	10
Internship	9	10
Course Project	9.29	10
National Event	9	10
Grand Total	43.38	100

5. Student Document Submission

Input:-

JSPM PRISM
STUDENT

Profile

Academic

Training and Placement

Submission

Attendance

CCE

Labs

Skills

Career & Placement

Feedback

Logout

JSPM's
Jayawantrao Sawant College of Engineering, Hadapsar, Pune.
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Achievements & Internships

Manage your certifications, internships, and project records.

Submission View History

Certification Internship Competition Field Project Mini Project

Edit Certification

Update the details below.

Course Name

C language

Organization Name

Aquadrove

Duration

2 months

Grade

-

Issue Date

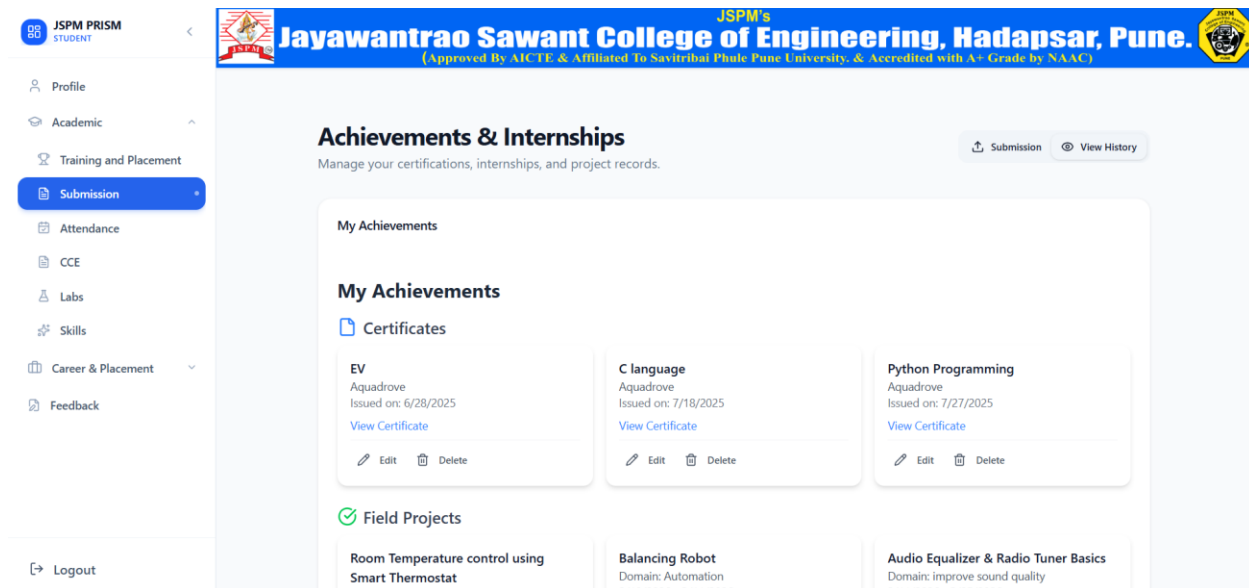
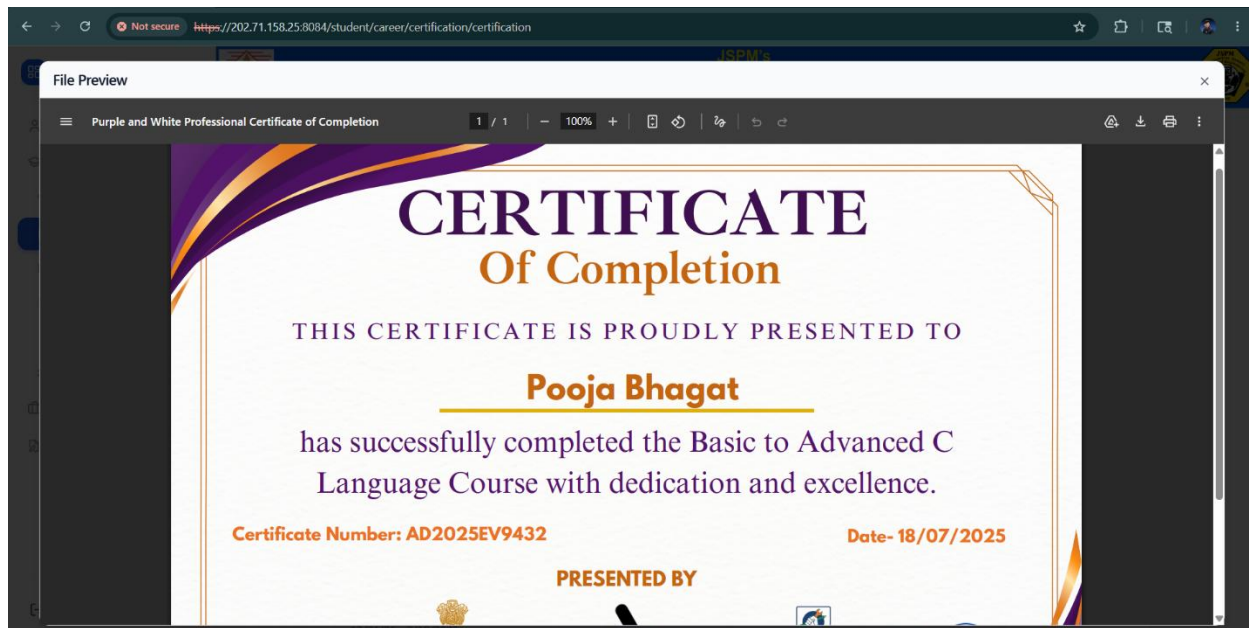
18-07-2025

Category

Technical

Documents (Optional if not changing)

Output:-



4. CODING

4.1 Code Snippets

Index.html

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <link rel="icon" type="image/svg+xml" href="/jspmicon.png" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>JSPM ERP</title>
  </head>
  <body>
    <div id="root"></div>
    <script type="module" src="/src/main.tsx"></script>
  </body>
</html>
```

App.jsx

```
import React, { Suspense, useEffect } from "react";
import { Outlet } from "react-router-dom";
import { Toaster, useToasterStore, toast } from "react-hot-toast";
import { Toaster as ShadcnToaster } from "@components/ui/toaster";
import Spinner from "@components/Spinner";
import GlobalModal from "@components/Common/GlobalModal";
```

```
export default function App() {
  const { toasts } = useToasterStore();
  const TOAST_LIMIT = 3;

  useEffect(() => {
    toasts
      .filter((t) => t.visible)
      .filter((_, i) => i >= TOAST_LIMIT)
      .forEach((t) => toast.dismiss(t.id));
  }, [toasts]);

  return (
    <ErrorBoundary>
      <Toaster position="top-center" />
      <ShadcnToaster />
      <GlobalModal />
      <Suspense fallback={<Spinner />}>
        <Outlet />
      </Suspense>
    </ErrorBoundary>
  );
}
```

```
}
```

Main.jsx

```
import { StrictMode } from "react";
import { createRoot } from "react-dom/client";
import { RouterProvider } from "react-router-dom";
import router from "@router";
import "./index.css";

createRoot(document.getElementById("root")!).render(
  <StrictMode>
    <RouterProvider router={router} />
  </StrictMode>
);
```

Routes.jsx

```
import { createBrowserRouter } from "react-router-dom";
import { lazy } from "react";
import App from "@App";
import {
  AdminRouteTree,
  StudentRouteTree,
  TeacherRouteTree,
} from "@router/lazyProtectedRoutes";
import authRoutes from "@router/routes/authRoutes";
import publicRoutes from "@router/routes/publicRoutes";

const NotFound = lazy(() => import("@pages/common/NotFound"));

const router = createBrowserRouter([
  {
    path: "/",
    element: <App />,
    children: [
      ...authRoutes,
      ...publicRoutes,
      { path: "student/*", element: <StudentRouteTree /> },
      { path: "admin/*", element: <AdminRouteTree /> },
      { path: "teacher/*", element: <TeacherRouteTree /> },
      { path: "*", element: <NotFound /> },
    ],
  },
]);

export default router;
```

PrivateRoute.jsx

```
import { useAtomValue, useSetAtom } from "jotai";
import { Navigate, Outlet } from "react-router-dom";
import { authAtom } from "@/State/Atom/authAtom";
import Cookies from "js-cookie";
import { useEffect } from "react";

interface PrivateRouteProps {
  requiredRole?: string | string[];
}

const PrivateRoute = ({ requiredRole }: PrivateRouteProps) => {
  const auth = useAtomValue(authAtom);
  const setAuth = useSetAtom(authAtom);

  const isDevBypass = import.meta.env.VITE_SHOW_DEV_TOOLS === "true";

  let token = Cookies.get("authToken");

  // keep token updated every 10 sec
  useEffect(() => {
    if (isDevBypass) return; // Skip interval if bypassing

    const interval = setInterval(() => {
      const newToken = Cookies.get("authToken");
      if (!newToken) {
        // token gone → user logged out
        localStorage.removeItem("auth");
        setAuth(undefined);
      }
    }, 10000);

    return () => clearInterval(interval);
  }, [setAuth, isDevBypass]);

  if (isDevBypass) {
    return <Outlet />;
  }

  token = Cookies.get("authToken");

  if (!token) {
    localStorage.removeItem("auth");
    setAuth(undefined);
    return <Navigate to="/" replace />;
  }
}
```

```

if (auth === undefined) {
  return <div>Loading...</div>;
}

if (!auth || !auth.user) {
  Cookies.remove("authToken");
  return <Navigate to="/" replace />;
}

const userRole = auth.user.role;

if (requiredRole) {
  const allowed = Array.isArray(requiredRole)
    ? requiredRole.includes(userRole)
    : userRole === requiredRole;

  if (!allowed) return <Navigate to="/" replace />;
}

return <Outlet />;
};

export default PrivateRoute;

```

Backend

AdmissionportalApplication.java

```

package in.edu.jspmjscoe.admissionportal;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class AdmissionportalApplication {

  public static void main(String[] args) {
    SpringApplication.run(AdmissionportalApplication.class, args);
  }
}

```

AuthController.java

```

package in.edu.jspmjscoe.admissionportal.controllers.auth;

import in.edu.jspmjscoe.admissionportal.dtos.apiresponse.ApiResponse;
import in.edu.jspmjscoe.admissionportal.model.security.password.ResetPasswordRequest;
import in.edu.jspmjscoe.admissionportal.model.teacher.Designation;

```

```
import in.edu.jspmjscoe.admissionportal.security.jwt.JwtUtils;
import in.edu.jspmjscoe.admissionportal.security.response.LoginResponse;
import in.edu.jspmjscoe.admissionportal.security.request.LoginRequest;
import in.edu.jspmjscoe.admissionportal.security.services.UserPrincipal;
import in.edu.jspmjscoe.admissionportal.services.security.UserService;
import lombok.RequiredArgsConstructor;
import io.swagger.v3.oas.annotations.tags.Tag;
import org.springframework.http.ResponseEntity;
import org.springframework.security.authentication.AuthenticationManager;
import org.springframework.security.authentication.UsernamePasswordAuthenticationToken;
import org.springframework.security.core.Authentication;
import org.springframework.security.core.context.SecurityContextHolder;
import org.springframework.web.bind.annotation.*;
```

```
@RestController
@RequestMapping("/api/auth/public")
@RequiredArgsConstructor
@Tag(name = "Auth Controller", description = "Authentication and Authorization endpoints")
public class AuthController {
```

```
    private final JwtUtils jwtUtils;
    private final AuthenticationManager authenticationManager;
    private final UserService userService;
```

```
    @PostMapping("/signin")
    public ResponseEntity<LoginResponse> authenticateUser(
        @RequestBody LoginRequest loginRequest) {
```

```
        Authentication authentication =
            authenticationManager.authenticate(
                new UsernamePasswordAuthenticationToken(
                    loginRequest.getUsername(),
                    loginRequest.getPassword()
                )
            );
```

```
        SecurityContextHolder.getContext().setAuthentication(authentication);
        userService.updateLoginDetails(loginRequest.getUsername());
        UserPrincipal principal = (UserPrincipal) authentication.getPrincipal();
```

```
        String accessToken = jwtUtils.generateToken(principal);
```

```
        return ResponseEntity.ok()
            .body(new LoginResponse(
                principal.getUsername(),
                principal.getAuthorities().stream()
                    .map(a -> a.getAuthority())
                    .toList(),
                accessToken
            ));
    }
```

```

        accessToken,
        principal.isFirstLogin(),
        principal.getDesignation(),
        principal.getYearOfStudy()
    ));
}

@PostMapping("/logout")
public ResponseEntity<?> logoutUser() {
    return ResponseEntity.ok()
        .body(new ApiResponse<>(200, "You've been signed out.", null));
}

@PostMapping("/forgot-password")
public ResponseEntity<String> forgotPassword(@RequestParam String email) {
    userService.forgotPassword(email);
    return ResponseEntity.ok("Reset link sent successfully");
}

@PostMapping("/reset-password")
public ResponseEntity<String> resetPassword(
    @RequestBody ResetPasswordRequest request) {
    userService.resetPassword(request);
    return ResponseEntity.ok("Password reset successful");
}

@GetMapping("/health")
public ResponseEntity<ApiResponse<String>> healthCheck() {
    return ResponseEntity.ok(new ApiResponse<>(200, "Backend is UP", null));
}
}

```

5. TESTING

5.1 Test Strategy

The testing strategy for the PRISM ERP system focuses on ensuring system security, reliability, functionality, and performance. Since the application handles sensitive academic and administrative data, testing was performed at multiple levels to verify both frontend and backend operations.

The following testing approaches were used:

5.1.1. Unit Testing

Individual modules such as authentication, profile management, marks entry, attendance, and document upload were tested independently to ensure correct functionality.

5.1.2. Integration Testing

Integration testing was performed to verify communication between:

- Frontend and backend APIs
- Spring Boot services and PostgreSQL database
- JWT authentication and authorization flow
- File upload service with MinIO storage

Because humans enjoy building systems where one module silently breaks another module at 2 AM. Integration testing exists purely because software engineers have trust issues with their own code. Correctly so.

5.1.3. System Testing

The complete ERP system was tested as a single integrated application to validate:

- User workflows
- Navigation
- Role-based access
- Dashboard operations
- Data consistency

5.1.4. Security Testing

Security testing was performed to ensure:

- Unauthorized users cannot access protected APIs
- JWT tokens are validated correctly
- Role-based restrictions are enforced
- Password-protected authentication works properly

5.1.5. User Acceptance Testing (UAT)

Faculty members and students tested the system using real academic workflows to verify usability and correctness.

5.2 Test Procedure

The following procedure was followed during testing of the PRISM ERP system:

Step 1: Requirement Verification

System requirements and module specifications were reviewed before testing.

Step 2: Environment Setup

Testing environment was configured using:

- Spring Boot backend
- PostgreSQL database
- React frontend
- Postman for API testing

Step 3: Module Testing

Each module was tested individually:

- Login module
- Student profile module
- Attendance module
- CCE marks module
- Placement module
- Document upload module

Step 4: API Testing

REST APIs were tested using Postman to verify:

- Request validation
- JWT authentication
- Response correctness
- Error handling

Step 5: Role-Based Access Testing

Testing was performed using multiple user roles:

- Admin
- HOD
- Teacher
- Student

Access permissions were verified for each role.

Step 6: Database Validation

Data entered through the frontend was verified directly in PostgreSQL tables to ensure integrity and consistency.

Step 7: Bug Fixing and Retesting

Detected issues were fixed and tested again to confirm successful resolution.

Because software testing is basically:

1. Break system
2. Fix system
3. Accidentally break another system
4. Pretend this is called “agile methodology”

5.3 Unit Test Case Plan

The Unit Testing phase was conducted to verify the correctness and functionality of individual modules of the PRISM ERP system independently before integration with other components.

The purpose of unit testing was to:

- Validate individual backend services and APIs
- Verify frontend component functionality
- Ensure proper database operations
- Detect logical and validation errors at an early stage

The following modules were included in unit testing:

Sr. No	Module Name	Testing Objective
1	Authentication Module	Verify login, JWT generation, and authorization
2	Student Profile Module	Validate profile creation and update operations
3	Attendance Module	Verify attendance entry and retrieval
4	CCE Management Module	Validate marks calculation and storage
5	Document Upload Module	Verify secure file upload and retrieval
6	Placement Module	Validate placement drive management
7	Resume Generation Module	Verify student resume generation
8	Role Management Module	Ensure proper access control enforcement

Tools Used for Unit Testing

- JUnit
- Postman
- Spring Boot Test
- Browser Developer Tools

Expected Outcomes

- All modules should function independently without errors
- APIs should return proper responses
- Database transactions should execute successfully
- Unauthorized access should be restricted

5.4 Acceptance Test Plan

Acceptance Testing was conducted to verify that the PRISM ERP system satisfies the academic and administrative requirements of the institution. The testing ensured that the developed system functions correctly in a real-world college environment and meets user expectations.

The Acceptance Test Plan focuses on validating:

- Functional correctness of the system
- Role-based access control
- User interface usability
- Data accuracy and consistency
- Security and authentication mechanisms

Objectives of Acceptance Testing

- Ensure all modules operate according to requirements
- Verify that users can perform assigned operations successfully
- Confirm secure access using JWT authentication
- Validate smooth interaction between frontend and backend
- Ensure system readiness for deployment

Users Involved in Acceptance Testing

Sr. No	User Role	Responsibility During Testing
1	Admin	Verify complete system administration
2	HOD	Verify department-level management features
3	Teacher	Verify marks entry and student management
4	Student	Verify profile access and academic viewing

Acceptance Testing Scope

The following modules were included in acceptance testing:

Sr. No	Module Name	Purpose of Testing
1	Authentication Module	Verify secure login and access control
2	Student Management Module	Validate student data operations
3	Attendance Module	Verify attendance management workflow
4	CCE Module	Validate marks entry and calculation

5	Document Management Module	Verify secure document uploads
6	Placement Module	Validate placement and T&P operations
7	Resume Module	Verify resume generation functionality

Acceptance Criteria

The system is considered accepted if:

- All major functionalities work correctly
- Users can access only authorized features
- Academic records are stored accurately
- Uploaded documents are securely managed
- System performance is stable during normal usage
- No critical security or functional defects remain

Expected Outcome

The PRISM ERP system should successfully support academic and administrative operations while maintaining security, accuracy, and usability for all authorized users.

5.5 Test Cases

Test Case ID	Pre Condition	Steps to Execute	Expected Result	Actual Result	Status	Priority
TC-Auth-01	Student account exists	Step 1: Open browser and navigate to application login page Step 2: Verify login form is displayed with Username and Password fields Step 3: Enter valid student username (e.g., student123) Step 4: Enter corresponding password Step 5: Click the Login button	User redirected to student dashboard	Student successfully logged in and redirected to Student Dashboard.	PASS	H
TC-Auth-02	Admin account exists	Step 1: Navigate to Login page Step 2: Clear any previous login fields Step 3: Enter valid admin username Step 4: Enter admin password Step 5: Click Login button	User redirected to admin dashboard	Admin successfully logged in and redirected to Admin Dashboard.	PASS	M
TC-Auth-03	Teacher account exists	Step 1: Go to Login page Step 2: Enter valid teacher username Step 3: Enter teacher password Step 4: Click Login button	User redirected to teacher dashboard	Teacher successfully logged in and redirected to Teacher Dashboard	PASS	L
TC-Auth-04	Login page loaded	Step 1: Enter non-existent username Step 2: Enter any password Step 3: Click Login button	Error: Invalid credentials displayed	System displayed "Invalid Credentials" error message for non-existent username.	PASS	H
TC-Auth-05	Valid user account exists	Step 1: Go to Login page Step 2: Enter valid username Step 3: Enter incorrect password Step 4: Click Login button	Error: Invalid credentials displayed	System displayed "Invalid Credentials" error message for incorrect password.	PASS	M

TC-Auth-06	Login page loaded	Step 1: Enter valid username Step 2: Enter incorrect password Step 3: Click Login button	User should not be logged in . An error message should be displayed indicating that the password must contain at least 8 characters .	User was logged in successfully and redirected to the dashboard despite entering a password with fewer than 8 characters.	FAIL	M
TC-CCE-01	Admin is logged in and on CCE page	1. Click Department dropdown 2. Select 'Electronics and Telecommunication' 3. Observe table data	Table should filter and display only students from the selected department	Department filter worked correctly. Only students from Electronics and Telecommunication department were displayed.	Pass	H
TC-CCE-02	Admin is logged in; Department selected	1. Click Batch dropdown 2. Select '2025-26' 3. Observe table	Table should display students of the selected batch only	Batch filter worked correctly. Records displayed only for the selected batch (2025–26).	Pass	H
TC-CCE-03	Admin is logged in; Department and Batch selected	1. Click Semester dropdown 2. Select 'Semester 4' 3. Observe table	Table should filter records for Semester 4 only	Semester filter worked correctly. Only Semester 4 records were displayed.	Pass	H
TC-CCE-04	Admin is logged in; Dept, Batch, Semester selected	1. Click Division dropdown 2. Select 'Division A' 3. Observe table	Table should display only Division A students	Division filter worked correctly. Only Division A students were displayed.	Pass	H
TC-CCE-05	Admin is logged in; all other filters selected	1. Click Subject dropdown 2. Select 'Communication' 3. Observe table	Table should display CCE marks for the selected subject	Subject filter worked correctly. Only students related to Communication subject were displayed.	Pass	H
TC-CCE-06	Admin is logged in on CCE page	1. Select a Department 2. Select a Batch with no enrolled students 3. Observe table	Table should show empty state message: 'No records found'	No records matched the selected filter combination. System displayed "No Records Found" message correctly.	Pass	M
TC-CCE-07	Admin is logged in on CCE page	1. Click Semester dropdown 2. Select 'Semester 4' 3. Observe table	Table should filter records for Semester 4 only without any error message	Semester 4 records were displayed successfully, but a "Server Error, Please Try Again Later" message was also shown.	FAIL	L
TC-Pro-01	Student is logged in and on Edit Profile page	1. Navigate to Edit Profile 2. Observe Candidate Name field	Candidate Name field should be pre-filled and non-editable (read-only)	Candidate Name field was pre-filled with existing profile data and displayed in read-only mode.	Pass	H

TC-Pro-02	Student is logged in and on Edit Profile page	1. Navigate to Edit Profile 2. Check Email Address field	Email field should show valid email: shrutib32005@gmail.com	Registered email address was displayed correctly and remained non-editable.	Pass	H
TC-Pro-03	Student is logged in and on Edit Profile page	1. Navigate to Edit Profile 2. Enter valid 10-digit mobile number: 9860497187 3. Click Save Changes	Mobile number should be accepted and saved successfully	Valid 10-digit mobile number was accepted and profile information was updated successfully.	Pass	H
TC-Pro-04	Student is logged in and on Edit Profile page	1. Navigate to Edit Profile 2. Clear mobile number field 3. Enter '98604' 4. Click Save Changes	Validation error should appear: 'Please enter a valid 10-digit mobile number'	System displayed validation message "Please enter a valid 10-digit mobile number" when invalid mobile number was entered.	Pass	H
TC-Pro-05	Student is logged in and on Edit Profile page	1. Navigate to Edit Profile 2. Click Gender dropdown 3. Select 'Female' 4. Click Save Changes	Gender should be saved as Female	Selected gender value (Female) was saved successfully and reflected after page refresh.	Pass	M
TC-Pro-06	Student is logged in and on Edit Profile page	1. Navigate to Edit Profile 2. Enter DOB: 03-07-2005 3. Click Save Changes	DOB should be saved correctly	Date of Birth was saved successfully and displayed correctly in the profile information.	Pass	M

5.6) Defect Report / Test Log

Defect Summary

Defect ID	Module	Defect Description	Severity
DEF-001	Authentication	Password validation not enforced correctly during login	Medium
DEF-002	CCE Management	Server error message displayed despite successful data retrieval	Low
DEF-003	Student Profile	Invalid mobile number accepted in certain scenarios	Medium
DEF-004	Training & Placement	Grand total score calculation mismatch for some students	High
DEF-005	Student Dashboard	Student records not refreshed after applying filters	Medium
DEF-006	Lab Management	Assigned batch not displayed immediately after assignment	Medium

DEF-007	Document Submission	PDF upload failed for files larger than permitted size	Low
DEF-008	Batch Management	Duplicate roll numbers allowed during batch creation	High

Test Log Summary

Module	Test Cases Executed	Passed	Failed	Status
Authentication & Authorization	15	14	1	Completed
Student Dashboard	28	27	1	Completed
Student Profile Management	25	25	0	Completed
CCE Management	30	28	2	Completed
Attendance Management	25	24	1	Completed
Training & Placement	28	27	1	Completed
Document Submission & Verification	25	24	1	Completed
Lab Management	25	24	1	Completed
Batch Management	25	25	0	Completed

Overall Testing Statistics

- Total Test Cases Executed: **226**
- Total Passed: **218**
- Total Failed: **8**
- Overall Pass Percentage: **96.46%**

6. USER MANUAL

Welcome to the PRISM Portal of JSPM JSCOE. This portal is used by Students, Teachers, HODs, and Admin staff to access academic and management services.

1. Login Page Instructions

1. Open the PRISM Login Page.
2. Enter your **Username**
 - Username should be your **registered Email ID**.
3. Enter your **Password**.
4. Click on the **Sign In** button.

User Login Details

User Type	Username	Password
Student	Registered Email ID	Student Password
Teacher	Registered Email ID	Teacher Password
HOD	Registered Email ID	HOD Password
Admin	Registered Email ID	Admin Password

Example:

- Username: example@jspmjscoe.edu.in (For Teacher)
- Password: Password@123

JSPM's Jayawantrao Sawant College of Engineering, Hadapsar, Pune.
(Approved By AICTE & Affiliated To Savitribai Phule Pune University, & Accredited with A+ Grade by NAAC)

JSPM - PRISM @ JSCOE

(Performance Review & Institutional Success Model)

A centralized platform to manage students, teachers, academics, and administration—designed to simplify college operations and improve communication across JSPM.

- Student Management**
Track attendance, results, profiles, and academic progress in real time.
- Faculty & Admin Control**
Manage teachers, subjects, schedules, notices, and internal workflows securely.

This system is developed and maintained by our dedicated contributors and technical team.

♥ Contributors

Welcome back
Sign in to continue

Username
rb0090120new@gmail.com

Password
.....

☐ Remember me for 30 days

Sign In

[Forgot your password?](#)
[Don't have an account? Create Account](#)

2. Student Panel

After successful login, the student is redirected to the **Student Profile Dashboard**

1. View Profile

1. After login, open **My Profile**.
2. View personal and academic information.

The screenshot displays the 'My Profile' page of the JSPM PRISM Student Portal. The header includes the college name 'Jayawantrao Sawant College of Engineering, Hadapsar, Pune.' and its affiliations. The left sidebar contains navigation links: Profile, Academic, Career & Placement, and Feedback. The main content area is titled 'Personal Profile Overview' and includes an 'Edit Profile' button. It features a student profile card for 'BHAGAT POOJA VIJAY' with a blue profile picture and 'Active Sem 4' status. Below the card, personal details are listed: Department (Electronics and Telecommunication...), Email Address (poojabhagat302006@gmail.com), Mobile Number (8999866704), and Aadhaar Card (**** 4545). The 'Personal Information' section contains a table with fields: Candidate Name, Current Semes..., Mobile Number, Birth Date, Religion, Department, Email Address, Gender, Caste Category, and Blood Group. The 'Contact Addresses' section shows 'CURRENT Address' (Sataw nagar handewadi road Hadapsar Pune) and 'PERMANENT Address' (Chanda Ghodegan Ahmednagar).

Personal Information	
Candidate Name	BHAGAT POOJA VIJAY
Current Semes...	4
Mobile Number	8999866704
Birth Date	31 May 2006
Religion	Hindu
Department	Electronics and Telecommunication...
Email Address	poojabhagat302006@gmail.com
Gender	Female
Caste Category	OBC
Blood Group	A+

Contact Addresses	
CURRENT Address	PERMANENT Address
Primary Info	Primary Info
Sataw nagar handewadi road Hadapsar Pune , Near Bharat petroleum	Chanda Ghodegan Ahmednagar , Near shanshignapur
Village/City	Village/City
Pune	AHMED NAGAR

Edit Profile

1. Click on **Edit Profile** button.
2. Update required details.
3. Click on **Save/Update** button.

The screenshot shows the 'Edit Personal Details' form in the JSPM PRISM Student Portal. The left sidebar is identical to the previous screenshot. The main content area is titled 'Edit Personal Details' and includes a 'Basic Information' section with fields for Candidate Name, Mobile Number, Date of Birth, Religion, Caste Category, Blood Group, Email Address, and Gender. The 'Current Address' section includes fields for Address Line 1, Address Line 2, State, District, Taluka, and City / Village. The student's profile card and personal details are also visible on the left.

Basic Information	
Candidate Name	BHAGAT POOJA VIJAY
Mobile Number *	8999866704
Date of Birth *	31-05-2006
Religion *	Hindu
Caste Category *	OBC (Other Backward Class)
Blood Group *	A+
Email Address *	poojabhagat302006@gmail.com
Gender *	Female

Current Address	
Address Line 1 *	Address Line 2
Sataw nagar handewadi road Hadap:	Near Bharat petroleum
State *	District *
Maharashtra	Ahmednagar
Taluka *	City / Village *
Nevasa	Pune

Download Resume

1. Open **My Profile** page.
2. Click on **Resume** button.
3. Resume will be generated automatically.
4. Click on **Download** to save the resume in PDF format.

The screenshot shows a student profile dashboard for Bhagat Pooja Vijay. At the top, there is a profile card with a blue circular profile picture placeholder, the name 'BHAGAT POOJA VIJAY', and contact information: 'Electronics and Telecommunication Engineering | Semester 4', 'poojabhagat302006@gmail.com | 8999866704', and 'Ahmednagar, Maharashtra | PRN: PRNENTC2102'. Below this, the 'PERSONAL DETAILS & INTERESTS' section contains a table with fields like Date of Birth, Gender, Department, Blood Group, Career Interest, Higher Studies, Religion, Category, Semester, and Placement. The 'EDUCATION' section features a table with columns for Qualification, Board/University, Year, and Percentage, listing 10th (SSC) and 12th (HSC) results.

Date of Birth	31 May 2006	Religion	Hindu
Gender	Female	Category	OBC
Department	Electronics and Telecommunication Engineering	Semester	4
Blood Group	A+	Placement	Yes
Career Interest	Embedded system		
Higher Studies	ME in electronic engineering		

Qualification	Board/University	Year	Percentage
10th (SSC)	MSBSHSE	2022	93%
12th (HSC)	MSBSHSE	2024	69%

2. Viewing Your Training & Placement Report

1. **Access the Page:** Click on **Training and Placement** from the left-hand menu.
2. **Verify Details:** Check the top section to ensure your Name, Roll No, and Division are correct.
3. **Review Scores:** Look at the table to see your marks broken down by **Employability** (Aptitude, Soft Skills) and **Technical Competency** (Projects, Internships, Certifications).
4. **Check Total:** Look at the bottom row to see your **Grand Total** score out of 100.

(Note: This is a read-only dashboard for you to track your overall performance).

- Profile
- Academic
- Training and Placement**
- Submission
- Attendance
- CCE
- Labs
- Skills
- Career & Placement
 - Industry Dashboard
 - Student Opportunities
 - My Applications
 - Feedback
- Logout

JSPM's Jayawantrao Sawant College of Engineering, Hadapsar, Pune.
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Student Training and Placement Report

Student Details
Name: BHAGAT POOJA VIJAY
Roll No: 2102
Division: A

Performance Overview (Total: 100 Marks)

Category	Score	Max
SPPU Result (SGPA)	0	20
Employability (30 Marks)		
Aptitude Test (Out of 10)	4.53	10
Softskill Test (Out of 10)	0.00	10
Softskill Attendance	0%	10
Technical Competency (50 Marks)		
Certification Courses	3	10
Subject Quiz	8.56	10
Internship	9	10
Course Project	9.29	10
National Event	9	10
Grand Total	43.38 / 100	

3. Document Submission and History

- Access the Page:** Click on **Submission** from the left-hand menu.
- Select Category:** Click the tab for the type of document you are adding (e.g., *Certification, Internship, Competition*).
- Fill in Details:** Enter the required information such as Course Name, Organization Name, Duration, and Grade in the provided text boxes.
- Upload Document:** Under the "Documents" section, click to upload your required **PDF** certificate or report (ensure it is under 10MB).
- Submit:** Click the **Submit** button at the bottom of the form to upload your record.
- View History:** To check the documents you have already submitted, click the **View History** button located near the top right of the page.

JSPM's

Jayawantrao Sawant College of Engineering, Hadapsar, Pune.

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Profile

Academic

Training and Placement

Submission

Attendance

CCE

Labs

Skills

Career & Placement

Feedback

Logout

Achievements & Internships

Manage your certifications, internships, and project records.

Submission

View History

Certification

Internship

Competition

Field Project

Mini Project

Certification Submission

Fill in the details below to add a new record.

Course Name

Enter course name

Organization Name

Enter organization name

Duration

e.g. 3 months

Grade

Grade obtained

Issue Date

dd-mm-yyyy

Category

Documents (PDF report is required)

4. Viewing CCE Marks

- Access the Page:** Click on **CCE** from the left-hand navigation menu.
- Verify Details:** Check the top section to ensure your Name, Roll No, and Division are correct.
- Select Category:** Click the tabs below your name (**Experiential Learning**, **Internal Test**, or **CCE Marks**) to switch between different evaluation types.
- Review Scores:** Look at the table to see your subject-by-subject breakdown (e.g., Field Activity, Quiz, Internal test, and Attendance).
- Check Total:** Look at the final column on the right to see your total calculated **CCE Marks** (out of 30) for each subject.

(Note: This page is "View Only," meaning it is automatically updated by your teachers for you to track your ongoing academic performance).

JSPM's

Jayawantrao Sawant College of Engineering, Hadapsar, Pune.

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Profile

Academic

Training and Placement

Submission

Attendance

CCE

Labs

Skills

Career & Placement

Industry Dashboard

Student Opportunities

My Applications

Logout

BHAGAT POOJA VIJAY

Roll No: 2102 Division: A

Subject Wise Marks (View Only)

Experiential Learning

Internal Test

CCE Marks

Subject Info	30 Marks Distribution				
Subject	Field Activity (15)	Quiz (5)	Internal test (5)	Attendance (5)	CCE Marks (30)
Communication Engineering	13.65	4.40	2.95	4.65	25.66
Control Systems	11.55	3.40	4.00	4.76	23.71
Environment Awareness	12.09	3.10	1.85	3.44	20.48
Object-Oriented Programming	12.60	4.63	3.40	4.24	24.88
Open Elective-II (Project Management)	9.45	4.00	1.80	3.46	18.71
Signals and Systems	12.75	1.87	3.85	4.61	23.07

2. Teacher panel:

1. Students Dashboard (HOD Access Only)

1. On the left-hand navigation menu, locate the **student** drop-down section.
2. Click on **Student Dashboard**.
3. **Find Students:** Use the dropdown filters at the top (Department, Batch, etc.) or type a name/roll number into the **Search records** bar.
4. **Add Students:** Click + **Add Student** to manually enter one student, or **Bulk Enroll** to upload a list at once.
5. **Reset Passwords:** Find the student in the table and click the padlock icon in the far-right **Reset Password** column.
6. **Export Data:** Click the **Download Excel** button in the top right to save your filtered student list as a spreadsheet.

JSPM PRISM ADMIN

Jayawantrao Sawant College of Engineering, Hadapsar, Pune.
(Approved by AICTE & Affiliated To Savitribai Phule Pune University, & Accredited with A+ Grade by NAAC)

Student Ledger
Central Management console

+ Add Student Bulk Enroll Download Excel

Department: Artificial Intelligence and Data Science Batch: 2025-26 Semester: Semester 3 Division: Division A

Search records...

Roll	Student Name	Institutional Email	Phone Number	Division	Reset Password
2101	THERESA MATTEL	theresamattel548@gmail.com	9028530390	A	🔒
2102	SHIVARKAR ANISHKA RAHUL	anishkashivarkar966@gmail.com	8530238676	A	🔒
2103	PATOLE NAMRATA SHASHIRAGANT	namratapatole361@gmail.com	8459467526	A	🔒
2104	GHOSH PRATHAMA ANISH	snigdha919@gmail.com	9903738307	A	🔒
2105	JORWAR PANKAJ RAJU	gorwarpankaj35@gmail.com	724894799	A	🔒
2106	NEMADE SIDDHESH TUSHAR	siddheshnemade27@gmail.com	8767321632	A	🔒
2107	SHINDE MONIKA SUBHASH	monika291006@gmail.com	9834686835	A	🔒
2108	MULE SUJAL SAMEER	sujalmule006@gmail.com	9322291913	A	🔒
2109	TONDARE SIYA MAHENDRA	siyatondare1326@gmail.com	7823027763	A	🔒
2110	PAWAR HARSHADA SACHIN	harshadapawar0973@gmail.com	9309114572	A	🔒

60 Total Records

Logout

2. Fill CCE Marks

1. **Navigate to CCE:** Click on **CCE** from the left-hand menu.
2. **Apply Filters:** At the top, select the correct *Department*, *Batch*, *Semester*, *Division*, and *Subject* from the dropdown menus.
3. **Select Category:** Click the appropriate tab for the marks you are entering (e.g., **CCE Marks**).
4. **Enter Marks:** In the student table below, locate the student and click into the text boxes (0) to enter their scores
(*Note: Totals calculate automatically*).
5. **Export Data:** Click the green **Download Excel** button in the top right to save a spreadsheet of the entered marks.

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COMPREHENSIVE CONTINUOUS EVALUATION
Manage and export student evaluation marks

Download Excel

Department: Artificial Intelligence Batch: 2025-26 Semester: Semester 4 Division: Division A Subject: Data Science

Experiential learning Internal test CCE Marks

STUDENT INFO		UNIT 1		UNIT 2		UNIT 3		UNIT 4		UNIT 5		TOTAL		OTHER
ROLL NO	NAME	FIELD (0-10)	QUIZ (0-10)	FIELD (0-10)	QUIZ (0-10)	FIELD (0-10)	QUIZ (0-10)	FIELD (0-10)	QUIZ (0-10)	FIELD (0-10)	QUIZ (0-10)	TOTAL QUIZ	TOTAL FIELD ACTIVITY	ATTENDANCE 100%
2101	THERESA MATTEL	4	10	9	10	10	10	9	10	9.5	10	50	41.5	89.29
2102	SHIVARKAR ANISHKA RAHUL	7.5	10	10	10	10	10	10	10	10	10	50	47.5	82.14
2103	PATOLE NAMRATA SHASHIKANT	7	10	8.5	10	10	10	9	10	9.5	10	50	44	89.29
2104	GHOSH PRATHAMA ANISH	5.5	10	8.5	10	6.5	10	9	10	2.5	10	50	32	60.71
2105	JORWAR PANKAJ RAJU	5	10	9	10	6.5	10	10	10	10	10	50	40.5	64.29
2106	NEMADE SIDDHESH TUSHAR	5.5	10	7	10	8.5	10	9.5	10	10	10	50	40.5	39.29
2107	SHINDE MONIKA	4	10	8.5	10	6.5	10	8.5	10	10	10	50	38.5	78.57

3. Training & Placement Marks Entry

- 1) **Access the Page:** On the left-hand menu, click on **Training & Placement** under the *student* section.
- 2) **Filter the Class:** At the top of the page, select the specific *Department*, *Batch*, *Semester*, and *Division* you want to assess.
- 3) **Choose the Evaluation Tab:** Select the appropriate category tab based on the activity you are grading:
 - a) **Employability (30%):** Click here to enter marks for Aptitude Tests (AT1-AT5), Soft Skills Tests (ST1-ST3), and Soft Skills Attendance.
 - b) **Technical Competency (50%):** Click here to enter marks related to technical skills and performance.
- 4) **Enter Student Marks:** Scroll down to the student table. Find the student by their Roll No or Name, and click directly into the empty text boxes (e.g., 0) to fill in their performance marks for the respective activities.
- 5) **Export Data:** The system saves your changes automatically. If you need a spreadsheet copy of these records, click the green **Download Excel** button in the top right corner.

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Training & Placement Management System
Comprehensive student performance tracking and assessment

All changes saved Download Excel

DepartmentBatchSemesterDivision

Artificial Intelligence and I+2025-26Semester 4Division A

Employability (30%)Technical Competency (50%)

STUDENT INFORMATION			MERIT ASSESSMENT (20%)	EMPLOYABILITY SKILLS (30%)								
ROLL NO	STUDENT NAME	DIV	SGPA SCORE (20 MARKS)	APTITUDE TESTS (MINIMUM 5 REQUIRED)					SOFT SKILLS TESTS (MINIMUM 3 REQUIRED)			SOFT SKILLS ATTENDANCE
				AT1	AT2	AT3	AT4	AT5	ST1	ST2	ST3	
2101	THERESA MATTEL	A	8.5	10	10	10	9.67	0	10	10	10	100 %
2102	SHIVARKAR ANISHKA RAHUL	A	8.45	0	0	0	0	0	10	10	10	100 %
2103	PATOLE NAMRATA SHASHIKANT	A	8.32	0	0	0	0	0	10	10	10	100 %
2104	GHOSH PRATHAMA ANISH	A	7.73	10	10	9.67	10	0	10	10	10	100 %
2105	JORWAR PANKAJ RAJU	A	8.64	10	10	8	7.67	6	0	0	0	0 %
2106	NEMADE SIDDHESH TUSHAR	A	8.27	10	10	8.33	9	8.33	0	0	0	0 %
2107	SHINDE MONIKA	A	8.23	10	10	10	9.67	0	10	10	10	100 %

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Training & Placement Management System
Comprehensive student performance tracking and assessment

All changes saved
Download Excel

Department: Artificial Intelligence and I+
Batch: 2025-26
Semester: Semester 4
Division: Division A
Employability (30%):
Technical Competency (50%):

STUDENT INFORMATION			TECHNICAL COMPETENCY ASSESSMENT (50%)					
ROLL NO	STUDENT NAME	DIV	CERTIFICATIONS	SUBJECT QUIZZES	INTERNSHIP	COURSE PROJECT	NATIONAL EVENTS	TOTAL SCORE
2101	THERESA MATTEL	A	2	11.48	0	10.56	0	60.47
2102	SHIVARKAR ANISHKA RAHUL	A	5	11.44	0	10.54	0	55.43
2103	PATOLE NAMRATA SHASHIKANT	A	2.67	11.48	0	10.56	0	53.03
2104	GHOSH PRATHAMA ANISH	A	4.4	10.86	0	9.26	0	60.18
2105	JORWAR PANKAJ RAJU	A	5	10.8	0	9.24	0	42.01
2106	NEMADE SIDDHESH TUSHAR	A	10	10.7	0	8.7	0	46.80
2107	SHINDE MONIKA SUBHASH	A	2.33	11.14	0	9.42	0	59.05
2108	MULE SUJAL SAMEER	A	1.8	10.88	0	8.96	0	38.14
2109	TONDARE SIYA MAHENDRA	A	0	11.36	0	10.2	0	40.38

4 Reviewing & Marking Student Submissions

- Access Submissions:** On the left-hand menu under the *student* section, click on **submission**.
- Select the Category & Class:** At the top of the page, use the dropdown filters to select the specific submission **Category** you want to review (e.g., *Competitions*, *Certifications*, *Internships*, etc.). Then, filter by *Department*, *Batch*, *Semester*, and *Division*.
- Review Documents:** In the table below, locate the student's submission. Click the green **View PDF** button in the *Document* column to verify the certificate or proof they uploaded.
- Assign Marks:** After reviewing the document, click directly into the text box in the **Marks** column (labeled e.g. 85/100) and type in the score the student has earned for that activity.
- Export Data:** To save a spreadsheet containing all the student details, submission names, and the marks you just entered, click the blue **Download as Excel** button.

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Category: Competitions
Department: Artificial Intelligence and Data Science
Batch: 2025-26
Semester: Sem 4
Division: Division A
Download as Excel

Student Name	Roll No	Competition Name	Organizer	Level	Award	Date	Document	Marks
THERESA MATTEL	2101	Buildathon	N/A	state	participant	2026-01-10	View PDF	e.g. 85/100
SHIVARKAR ANISHKA RAHUL	2102	Group Discussion Competition	N/A	institutional	3rd-prize	2025-09-29	View PDF	e.g. 85/100
		COGNI AI Poster Presentation Competition	N/A	institutional	1st-prize	2026-02-28	View PDF	e.g. 85/100
PATOLE	2103	Cogni AI Forum						

5. Master Batch Setup (HOD Access Only)

1. **Navigate to Batch:** Click on **Batch** under the *student* section from the left-hand menu.
2. **Apply Filters:** At the top, select the correct Department, Batch, Semester, and Division from the dropdown menus.
3. **Select Tab:** Click on the **Create New Batch** tab to start forming a group.
4. **Add Students:** Enter a **Batch Name** (e.g., A1). Then, add students by either entering a **ROLL NO RANGE** and clicking **Apply Range**, or by manually checking the boxes next to their names in the list.
5. **Finalize:** Click the purple **Finalize Batch** button near the top right to save the newly created student group.
6. **View Existing Batches:** Click on the **Existing Batches** tab to view or manage the student groups you have already created.

The screenshot displays the 'Master Batch Setup' interface within the JSPM PRISM Admin system. The header includes the JSPM logo and the college name: 'Jayawantrao Sawant College of Engineering, Hadapsar, Pune.' Below the header, a left-hand navigation menu lists various options, with 'Batch' currently selected. The main content area is titled 'Master Batch Setup' and includes filters for Department (Artificial Intelligence and Data Science), Batch (2025-26), Semester (Semester 3), and Division (Division A). There are two tabs: 'Existing Batches' and 'Create New Batch', with the latter being active. The 'Create New Batch' section contains a 'Batch Name (e.g. A1)' input field and a 'ROLL NO RANGE' section with 'Start' and 'End' input fields and an 'Apply Range' button. A note states: 'Note: Students selected here will be assigned to batch'. To the right, the 'Student List' section shows a table of students with checkboxes for selection. The table lists students with their roll numbers and names: 2114 SHUKLA HITESH JAYESH, 2115 GOLE ANUSHKA MAHADEV, 2116 KORE GAYATRI VIJAYKUMAR, 2117 SHINDE OMKAR HARISH, 2118 BHOSALE SUSHANT SOMNATH, 2119 MAINDAD OMKAR DHANANJAY, 2120 PUPPALWAR SOHAM PANDHARI, and 2121 DHAGE GAURAV RAVINDRA. At the top right of the student list, it says 'SELECTED: 0 STUDENTS' and there is a 'Finalize Batch' button.

Roll No	Student Name	Select
2114	SHUKLA HITESH JAYESH	☐
2115	GOLE ANUSHKA MAHADEV	☐
2116	KORE GAYATRI VIJAYKUMAR	☐
2117	SHINDE OMKAR HARISH	☐
2118	BHOSALE SUSHANT SOMNATH	☐
2119	MAINDAD OMKAR DHANANJAY	☐
2120	PUPPALWAR SOHAM PANDHARI	☐
2121	DHAGE GAURAV RAVINDRA	☐

6. Assigning Labs to Teachers with batches (HOD Access Only)

1. **Navigate to Lab Management:** Click on **Lab Management** from the left-hand menu.
2. **Apply Filters:** At the top, select the correct Department, Batch, Semester, Division, and Subject from the dropdown menus.
3. **Select the Lab:** Click on the specific laboratory from the list on the left side (e.g., *Artificial Intelligence Laboratory*).
4. **Assign the Batch:** Click the blue **+ Assign Batch** button on the right side of the screen.
5. **Select Teacher & Batch:** In the pop-up or designated area, select the specific **Teacher** and the student **Batch** (e.g., A1) you want to link together for this lab, then save.
6. **Verify Assignment:** Scroll down to the **ASSIGNED BATCHES** table at the bottom to confirm that the teacher and batch are successfully linked to the lab.

(Optional: You can also use the **Add Experiments** button to define the specific experiments for the lab syllabus).

7. LIMITATIONS / DRAWBACKS OF PROPOSED SYSTEM

Although the PRISM ERP system provides a secure and centralized academic management platform, the current version has certain limitations:

- The system is primarily designed for college-level academic operations and may require customization for other institutions.
- Internet connectivity is required for accessing the web application.
- The system currently supports only predefined user roles such as Admin, HOD, Teacher, and Student.
- Mobile application support is not available in the current version.
- Advanced analytics and AI-based student performance prediction features are not implemented.
- Large-scale concurrent access may require additional server optimization and load balancing.
- Offline functionality is not supported.
- Integration with external university ERP systems or government portals is currently unavailable.

8. FUTURE ENHANCEMENT

The following enhancements can be implemented in future versions of the PRISM ERP system:

- Development of Android and iOS mobile applications.
 - Integration with university examination portals.
 - AI-based student performance analysis and prediction.
 - Real-time notification system using Email and SMS.
 - Online examination and quiz management module.
 - Biometric attendance integration.
 - Cloud deployment for improved scalability and availability.
 - Advanced reporting and analytics dashboard.
 - Video lecture and learning management integration.
 - Multi-college or multi-campus support.
 - Automated timetable generation system.
 - Integration with payment gateways for fee management.
-

9. CONCLUSION

The PRISM ERP system successfully provides a secure, centralized, and role-based platform for managing academic and administrative activities within an educational institution.

The system improves efficiency by digitizing processes such as:

- Student management
- Attendance tracking
- CCE marks management
- Document handling
- Training and placement activities

By implementing JWT-based authentication and role-based authorization, the system ensures secure access control and protects sensitive academic data.

The modular architecture of the system makes it scalable, maintainable, and suitable for future enhancements. The project demonstrates the effective use of modern web technologies such as Spring Boot, React.js, PostgreSQL, and REST APIs in developing an enterprise-level academic management solution.

Overall, the PRISM ERP system reduces manual workload, improves data accuracy, enhances transparency, and provides a strong foundation for digital academic management.

10. BIBLIOGRAPHY

1. Craig Walls, *Spring Boot in Action*, Manning Publications, 2016.
2. Laurentiu Spilca, *Spring Security in Action*, Manning Publications, 2020.
3. Roger S. Pressman and Bruce R. Maxim, *Software Engineering: A Practitioner's Approach*, 9th Edition, McGraw-Hill Education, 2019.
4. Ian Sommerville, *Software Engineering*, 10th Edition, Pearson Education, 2015.
5. Oracle Corporation, *Java Platform Documentation*. Available at:
<https://docs.oracle.com/en/java/>
6. Spring Boot Official Documentation. Available at: <https://spring.io/projects/spring-boot>
7. Spring Security Official Documentation. Available at: <https://spring.io/projects/spring-security>
8. PostgreSQL Official Documentation. Available at: <https://www.postgresql.org/docs/>
9. React Official Documentation. Available at: <https://react.dev/>
10. JWT (JSON Web Token) Official Documentation. Available at: <https://jwt.io/>
11. Docker Official Documentation. Available at: <https://docs.docker.com/>
12. Nigel Poulton, *Docker Deep Dive*, Independently Published.
13. Postman Learning Center Documentation. Available at: <https://learning.postman.com/>
14. Selenium Official Documentation. Available at:
<https://www.selenium.dev/documentation/>
15. Unmesh Gundecha, *Selenium Testing Tools Cookbook*, Packt Publishing.
16. JUnit 5 User Guide. Available at: <https://junit.org/junit5/docs/current/user-guide/>
17. Apache Maven Documentation. Available at: <https://maven.apache.org/guides/>
18. OpenAPI (Swagger) Documentation. Available at: <https://swagger.io/docs/>
19. IntelliJ IDEA Documentation. Available at:
<https://www.jetbrains.com/idea/documentation/>
20. Git Documentation. Available at: <https://git-scm.com/doc>
21. GitHub Documentation. Available at: <https://docs.github.com/>
22. Savitribai Phule Pune University (SPPU), MCA Project Guidelines and Academic Documentation Standards.
23. JSPM's Jayawantrao Sawant College of Engineering, Internal Project Development Guidelines and ERP Requirements Documents.